MBG 1201 Introduction to Modern Biology

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Outline

Discovery of the nature of the gene
Structure Of The DNA
DNA replication
Nucleotide Excision Repair of DNA
Introduction to Gene Expression

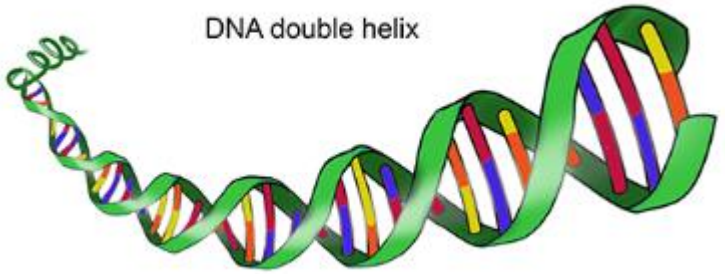
What are genes made of?

T. H. Morgan's group showed that genes are located on chromosomes and that the behavior of chromosomes during meiosis accounts for Mendel's laws.

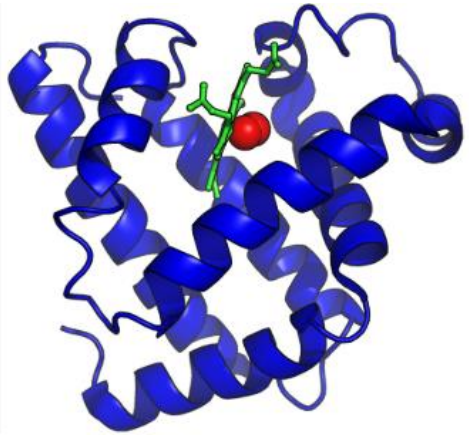
The two components of chromosomes — DNA and protein — became candidates for the genetic material.

- The key factor in determining the genetic material was choosing appropriate experimental organisms.

- The role of DNA in heredity was first discovered by studying bacteria and the viruses that infect them.



vs



Evidence That DNA Is The Genetic Material

First evidence: Genes Can Transform Bacteria

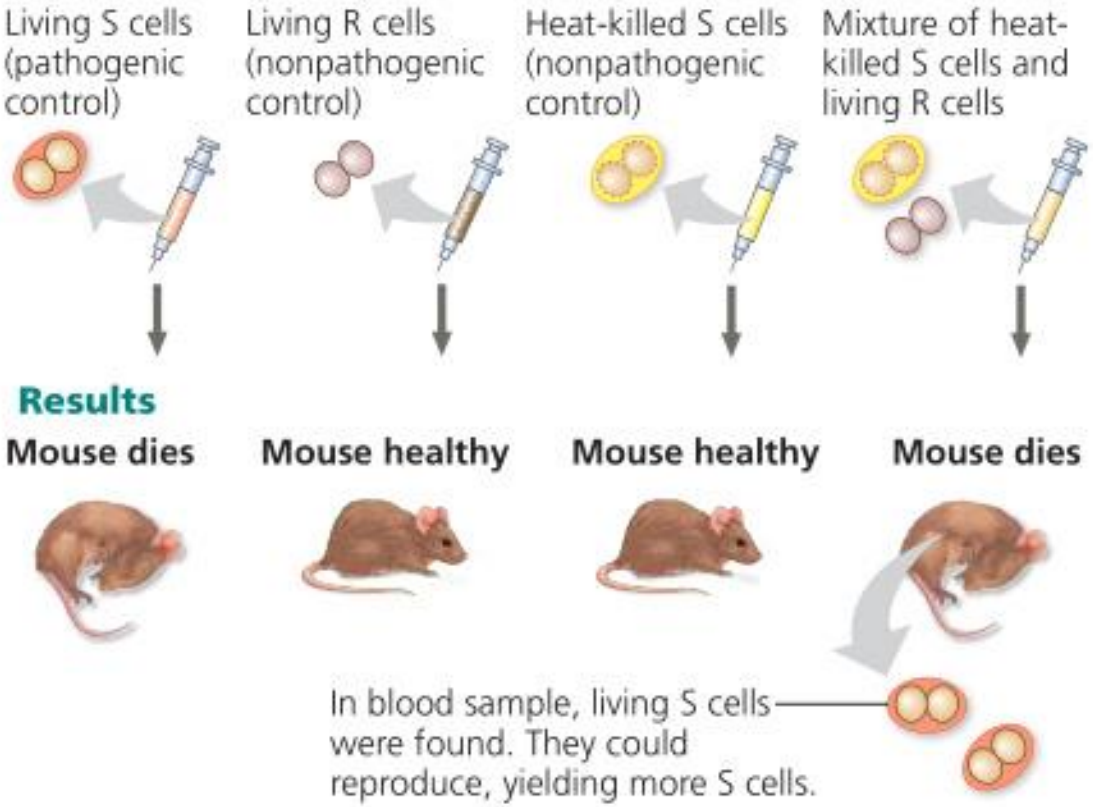
Frederick Griffith's research on Streptococcus pneumoniae (bacteria that cause pneumonia) in 1928 while he was trying to develop a vaccine for pneumonia.

Griffith experimented with two strains of S. pneumoniae:
(Strain: a genetic variant within a biological species.)

R (Rough) strain: non-pathogenic (harmless) strain does not cause disease when injected into mouse
S (Smooth) strain: pathogenic (disease causing) strain cause disease when injected into mouse
Heat killed S cells does not cause disease when injected into mouse

Can a genetic trait be transferred between different bacterial strains?

Experiment Frederick Griffith studied two strains of the bacterium *Streptococcus pneumoniae*. The S (smooth) strain can cause pneumonia in mice; it is pathogenic because the cells have an outer capsule that protects them from an animal's immune system. Cells of the R (rough) strain lack a capsule and are nonpathogenic. To test for the trait of pathogenicity, Griffith injected mice with the two strains:



Conclusion The living R bacteria had been transformed into pathogenic S bacteria by an unknown, heritable substance from the dead S cells that enabled the R cells to make capsules.

Griffith mixed heat-killed S cells with live R cells and injected the mixture into a mouse.
The mouse died and Griffith found live S cells in its blood.
Furthermore, S cells recovered from blood of the dead mouse inherited the trait of pathogenicity to their offspring.
He called this phenomenon **transformation** which is described as a change in genotype and phenotype due to the assimilation of external DNA by a cell.

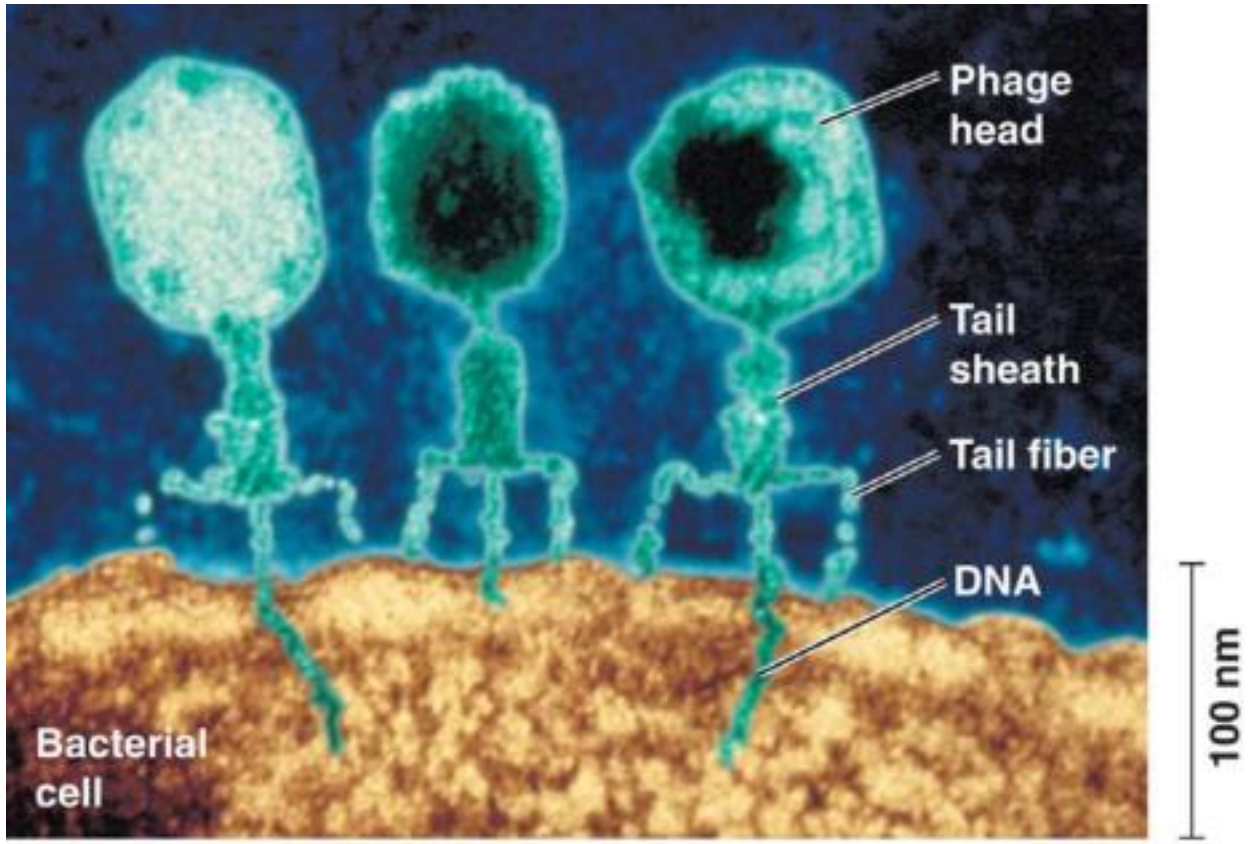
Griffith reasoned that even if s cells were dead their genetic material was taken up by r cells and this transformed them into pathogenic s cells.

Later work by Oswald Avery, Maclyn McCarty, and Colin MacLeod identified the transforming substance as DNA (1944). They isolated DNA from the heat killed S strain. When the material wasdigested with enzymes which break down proteins or RNA it was still pathogenic. When it was digested with enzymes which break down DNA it was not pathogenic.

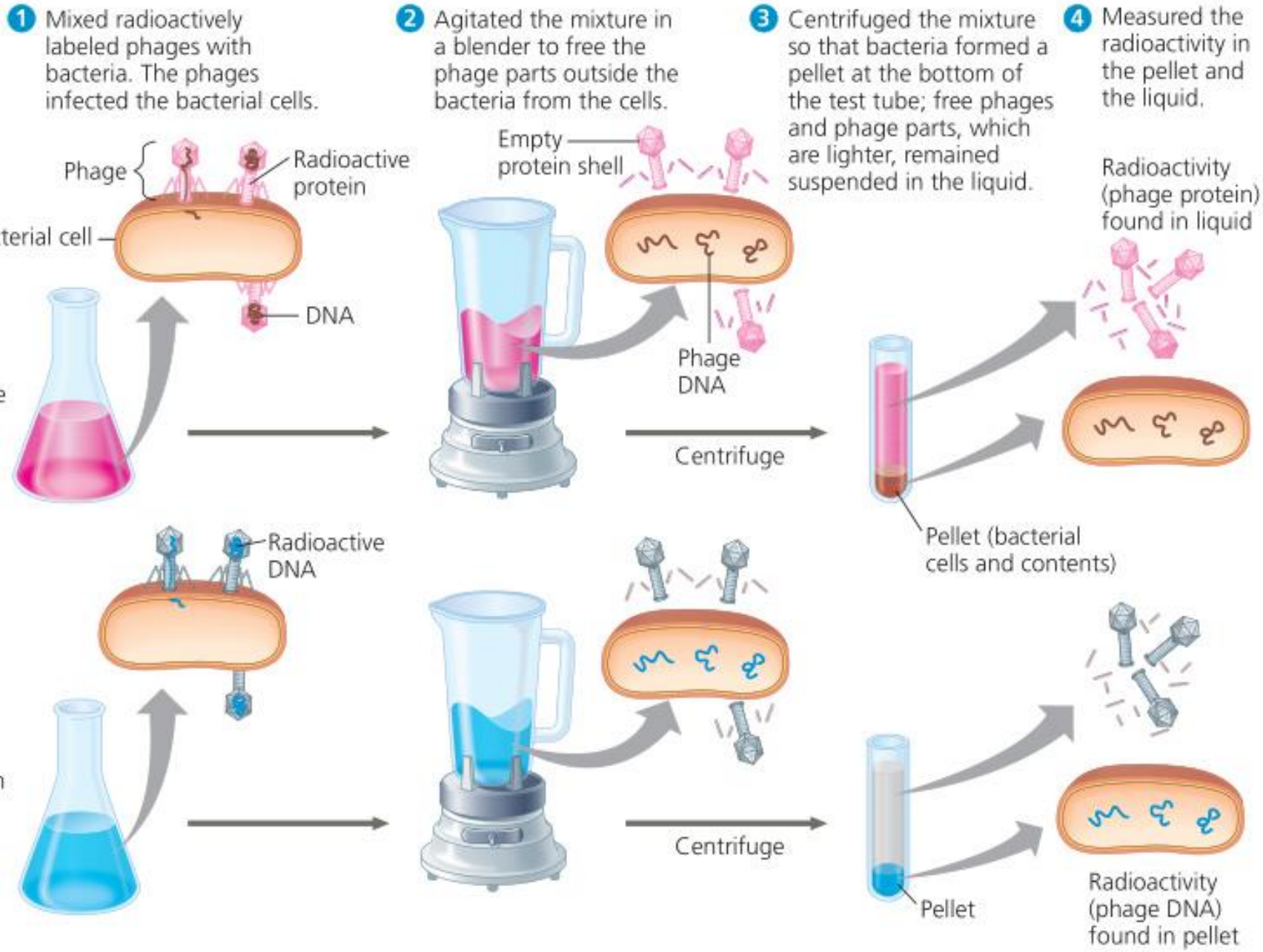
Evidence That DNA Is The Genetic Material

Second Evidence: Viral DNA Can Program Cells

Viruses are composed of DNA/RNA and protein.
Bacteriophages are viruses that infect bacteria.
Since viruses don't have a cell, To produce more viruses, a virus must infect a cell and take over the cell's metabolic machinery.
In 1952, Alfred Hershey and Martha Chase performed experiments showing that DNA is the genetic material of a phage known as T2.
T2 attaches to the cell surface of E. coli (bacteria) and injects its genetic material inside the bacteria while its protein coat remains outside the cell.
To determine the source of genetic material in the phage, they designed an experiment showing that only one of the two components of T2 (DNA or protein) enters an E. coli cell during infection.



Experiment Alfred Hershey and Martha Chase used radioactive sulfur and phosphorus to trace the fates of protein and DNA, respectively, of T2 phages that infected bacterial cells. They wanted to see which of these molecules entered the cells and could reprogram them to make more phages.



Batch 1: Phages were grown with radioactive sulfur (^{35}S), which was incorporated into phage protein (pink).

Batch 2: Phages were grown with radioactive phosphorus (^{32}P), which was incorporated into phage DNA (blue).

Results: When proteins were labeled (batch 1), radioactivity remained outside the cells, but when DNA was labeled (batch 2), radioactivity was found inside the cells. Cells containing radioactive phage DNA released new phages with some radioactive phosphorus.

Conclusion Phage DNA entered bacterial cells, but phage proteins did not. **Hershey and Chase concluded that DNA, not protein, functions as the genetic material of phage T2.**

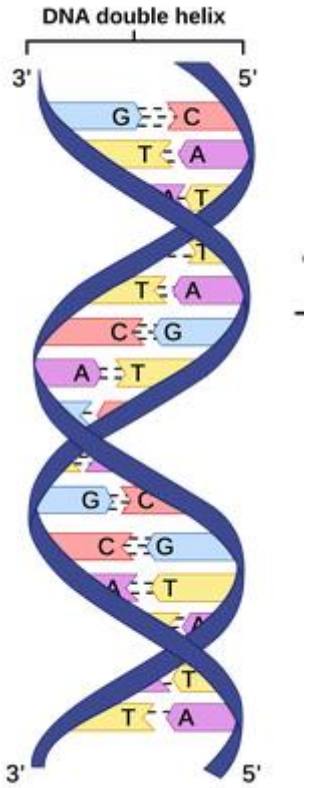
Hershey/Chase Experiments

• They grew one batch of T2 phage in the presence of radioactive sulfur, marking the proteins (but not the DNA) radioactive.
• They grew another batch in the presence of radioactive phosphorus, marking the DNA (but not the proteins) radioactive.
• They allowed each batch to infect separate E. coli cultures.
• Shortly after the onset of infection, Hershey and Chase spun the cultured infected cells in a blender, shaking loose any parts of the phage that remained outside the bacteria.
• The mixtures were spun in a centrifuge, which separated the heavier bacterial cells in the pellet from the lighter free phages and parts of phages in the liquid supernatant.
• They then tested the pellet and supernatant of the separate treatments for the presence of radioactivity.
– The bacteria infected with T2 phages containing radiolabeled proteins: most of the radioactivity was in the supernatant that contained phage particles, not in the pellet with the bacteria.
– The bacteria infected with T2 phages containing radiolabeled DNA: most of the radioactivity was in the pellet with the bacteria.

They concluded that the injected DNA of the phage provides the genetic information that makes the infected cells produce new viral DNA and proteins to assemble into new viruses.

Additional Evidence from Chargaff's experiments

It was known that DNA is a polymer of nucleotides, each consisting of one of the four types of nitrogen containing base (adenine. guanine, cytosine or thymine) , a deoxyribose sugar, and a phosphate group.



- In 1950, biochemist Erwin Chargaff reported that DNA composition varies from one species to the next.

For example, he found that 32.8% of sea urchin DNA nucleotides have the base A, whereas only 24.7% of those from the bacterium E. coli have the base A.

- This evidence of diversity made DNA a more credible candidate for the genetic material.
- In the DNA of each species he studied, the number of adenines approximately equaled the number of thymines, and the number of guanines approximately equaled the number of cytosines.

For example In sea urchin DNA the four bases are present in these percentages:

A = 32.8% and T = 32.1%; G = 17.7% and C = 17.3%.

Chargaff’s rules

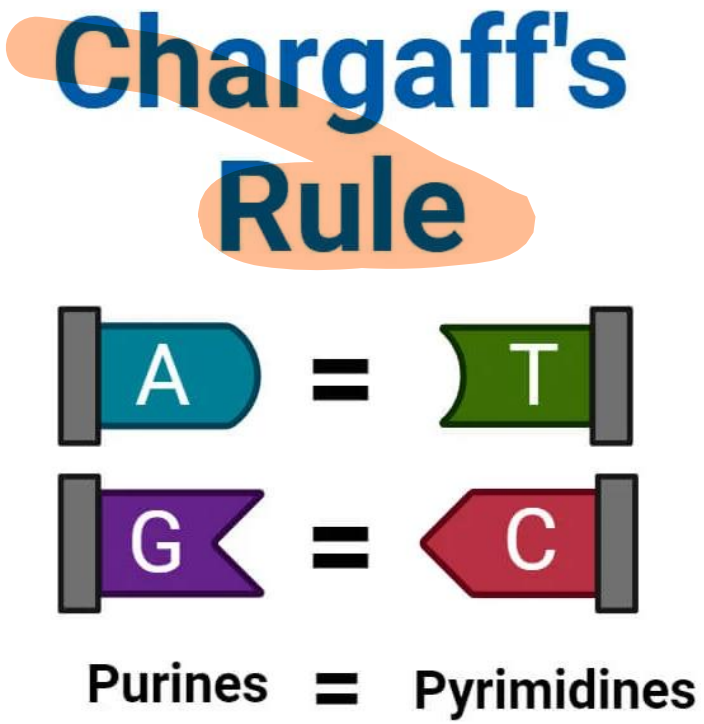
Chargaff’s 2 findings became known as Chargaff’s rules

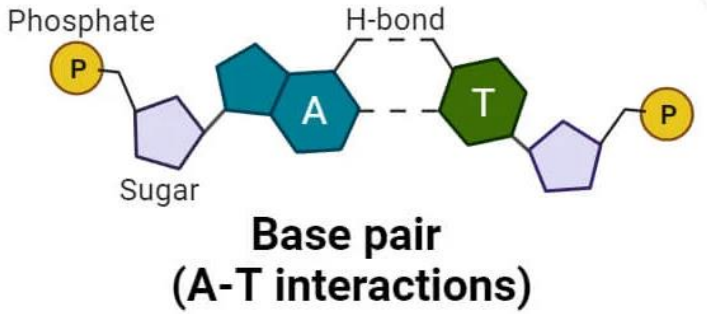
1)The base composition of DNA varies between species

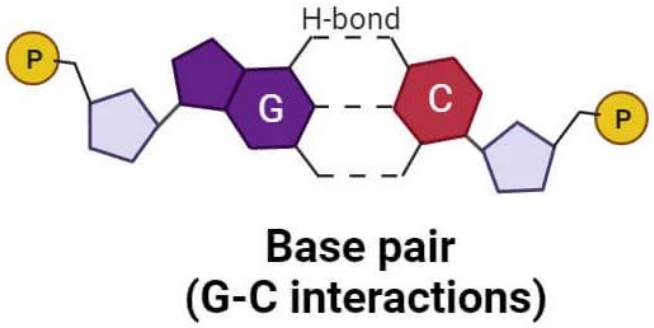
2) In any species

number of A = number of T

number of G = number of C







Building a Structural Model of DNA

After most biologists became convinced that DNA was the genetic material, the challenge was to determine how its 3-D structure accounts for its role in heredity.

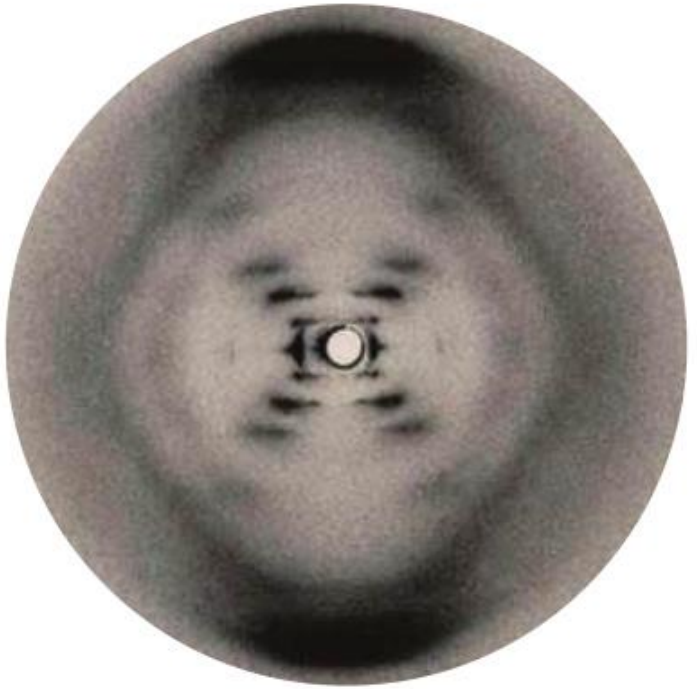
- Maurice Wilkins and Rosalind Franklin were using a technique called X-ray crystallography to study molecular structure.

- Franklin produced pictures of the DNA molecule using this technique.

In her unpublished work, she concluded that there were two antiparallel sugar-phosphate backbones, with the nitrogenous Bases paired in the molecule's interior.



(a) Rosalind Franklin

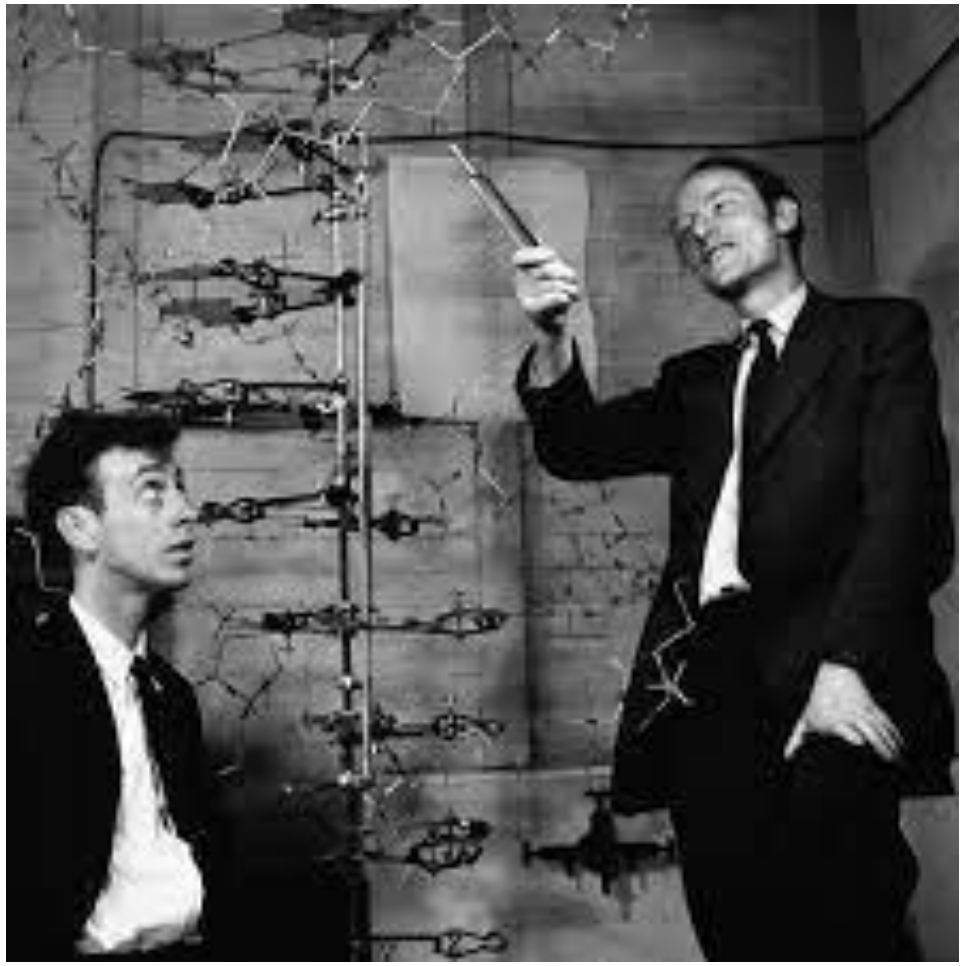


(b) Franklin's X-ray diffraction photograph of DNA

Building a Structural Model of DNA

• Franklin's X-ray crystallographic images and unpublished work enabled James Watson and Francis Crick to deduce the first accurate model of DNA.

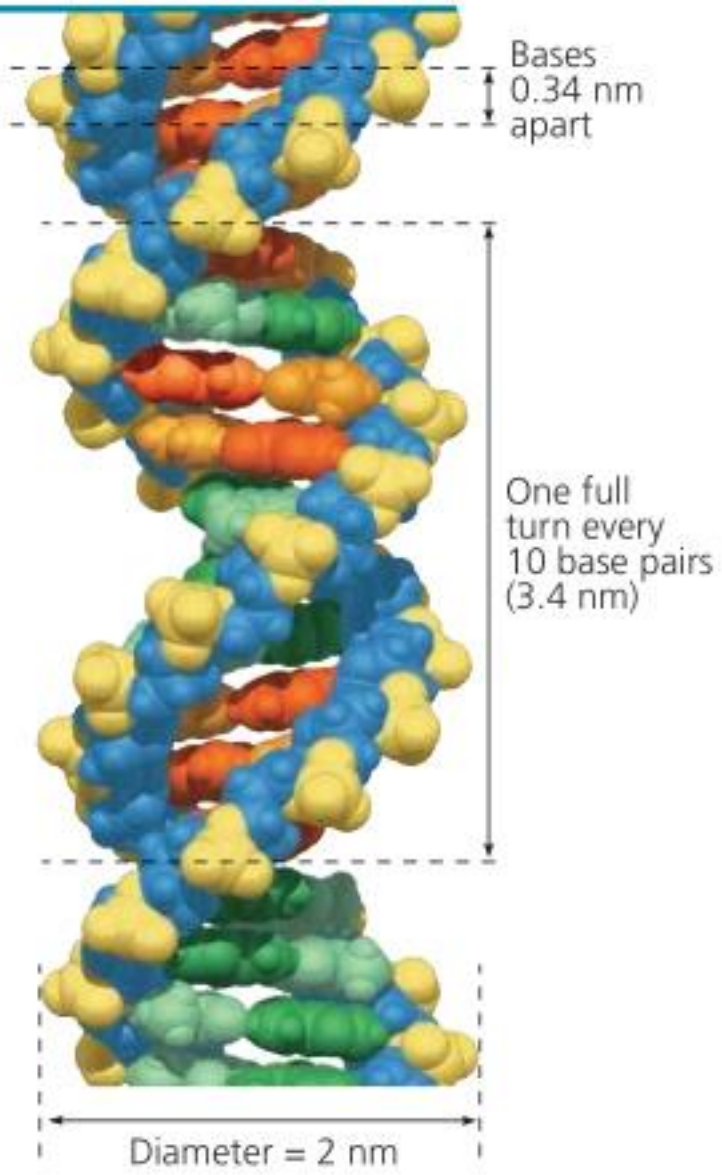
Rosalind Franklin's DNA picture was
the basis of the discovery of DNA structure
which was awarded with a Nobel prize in 1962.
But since Franklin died in 1958 she was not given
a Nobel Prize.

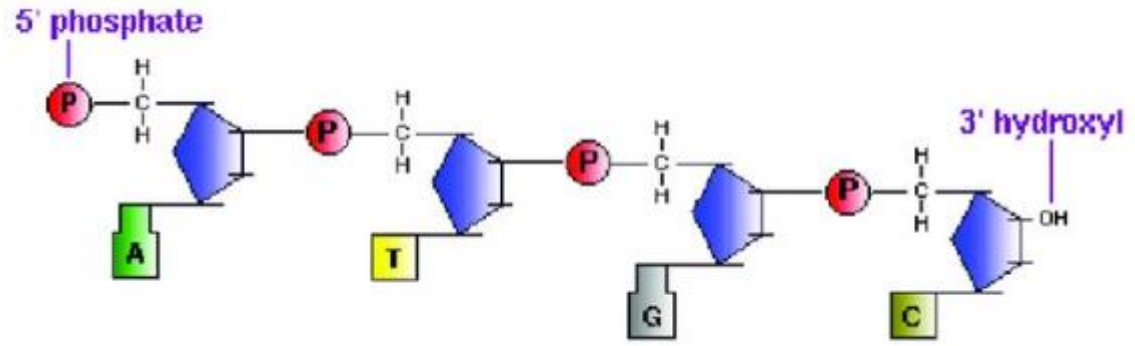


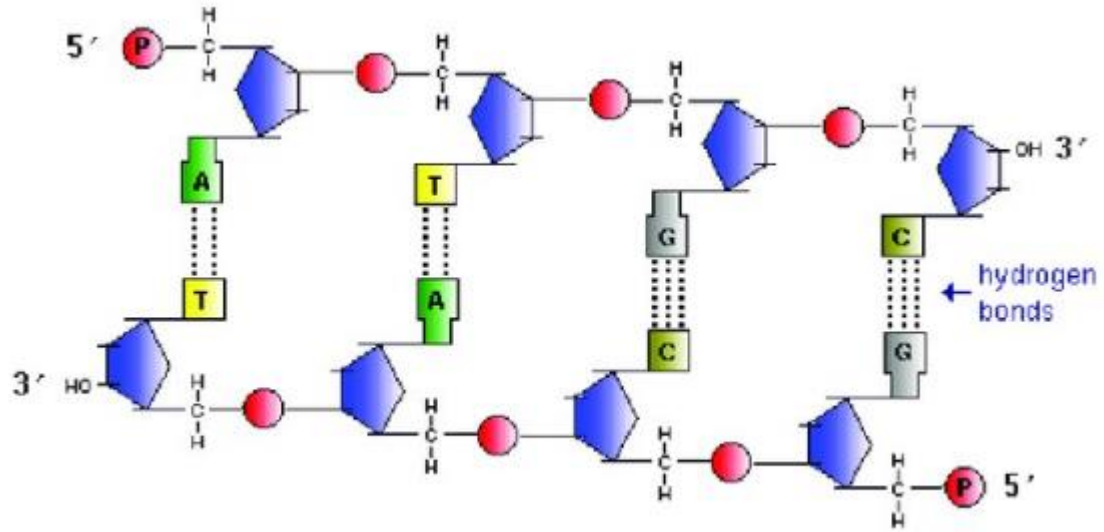
Building a Structural Model of DNA

Structural Images

The space-filling model on the left shows the 3-D shape of the DNA double helix; the diagram on the right shows chemical details of DNA's structure. In both images, phosphate groups are yellow, deoxyribose sugars are blue, and nitrogenous bases are shades of green and orange.







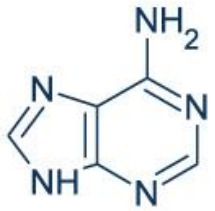
The DNA double helix is "right-handed." Use your right hand as shown to follow the sugar-phosphate backbone up the helix and around to the back. (It won't work with your left hand.)



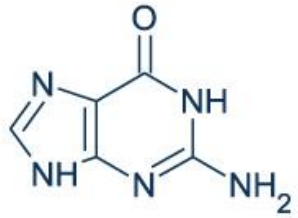
Notice the uniform diameter.

Base Pairing in DNA

Purines

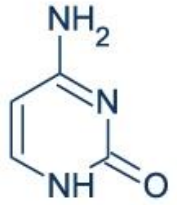


Adenine

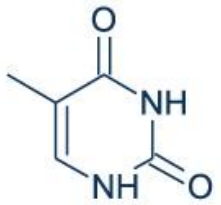


Guanine

Pyrimidines



Cytosine

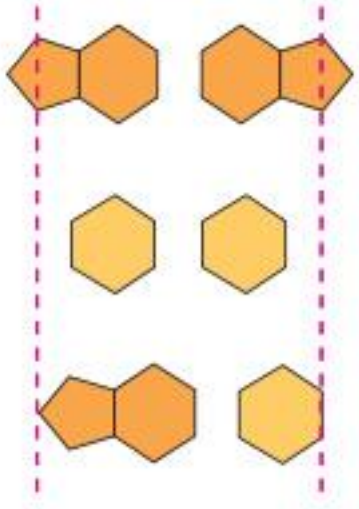


Thymine

Adenine and guanine are **purines** (2 rings)
cytosine and thymine are pyrimidines (have 1 ring).

Pairing a purine with a pyrimidine is the only combination that results in a uniform diameter (2nm) for the double helix.

▼ **Figure 16.8** **Possible base pairings in the DNA double helix.**



Purine + purine: too wide

Pyrimidine + pyrimidine: too narrow

Purine + pyrimidine: width consistent with X-ray data

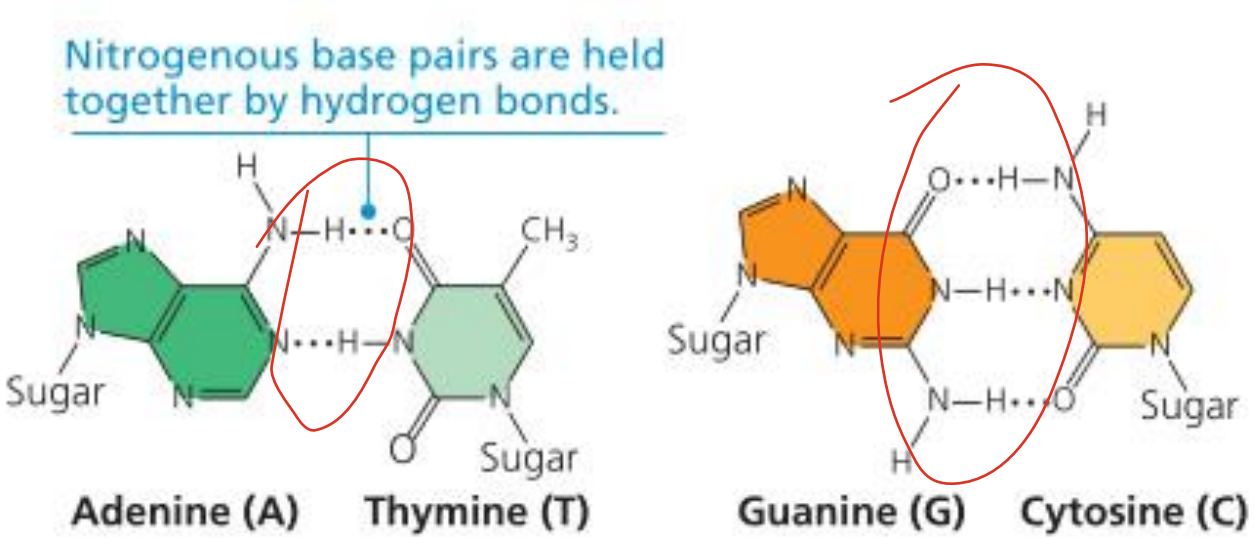
Base Pairing in DNA

Each base has chemical side groups that can form hydrogen bonds with its appropriate partner: Adenine can form two hydrogen bonds with thymine and only thymine; guanine forms three hydrogen bonds with cytosine and only cytosine.

Wherever one strand of a DNA molecule has an A, the partner strand has a T. Similarly, a G in one strand is always paired with a C in the complementary strand.

A pairs with T, and G pairs with C

The Watson-Crick explains Chargaff's rule regarding the equal ratios of adenines to thymines and guanines to cytosines in any organism.



Human DNA is 30.3% adenine, 30.3% thymine, 19.5% guanine, and 19.9% cytosine.

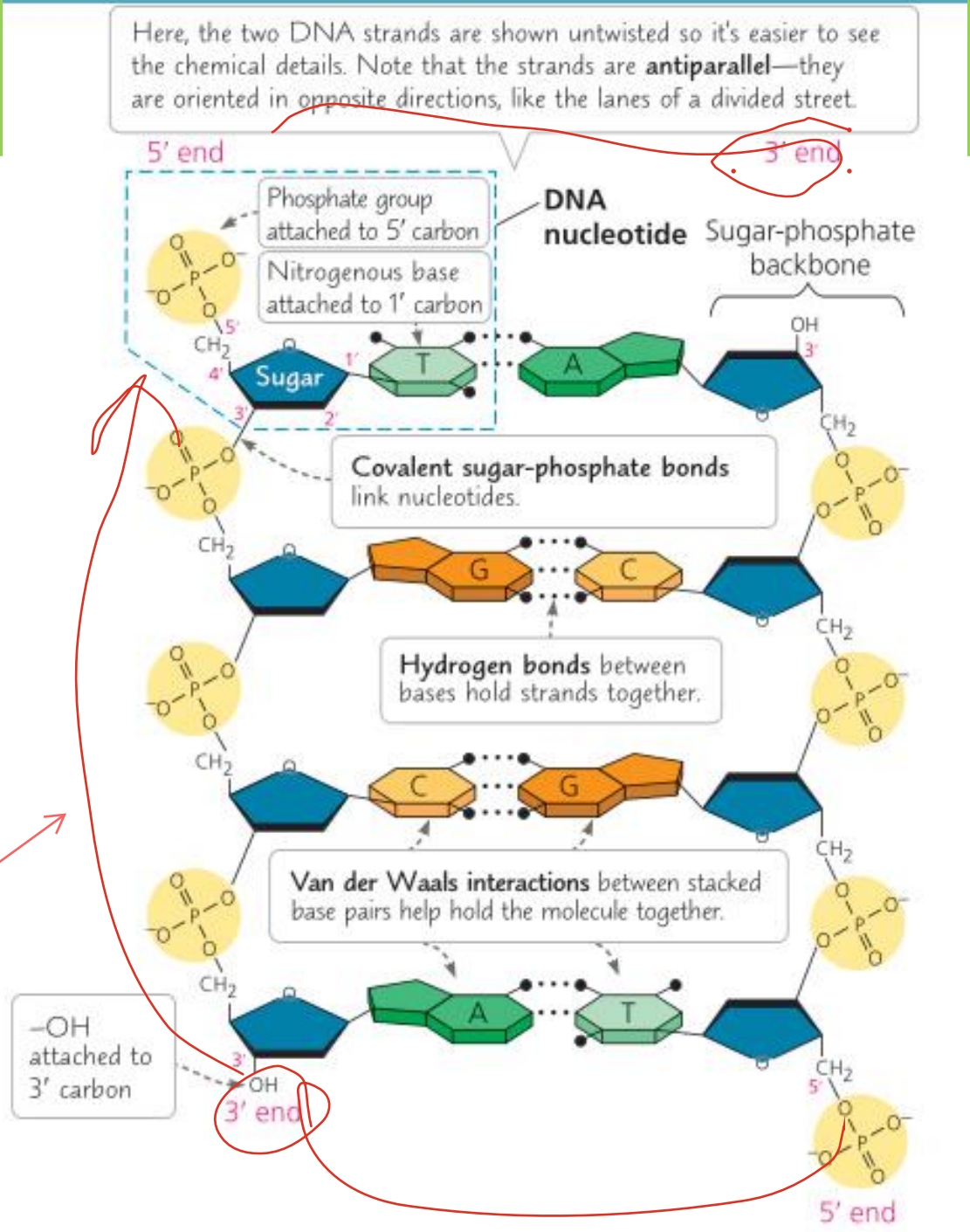
Base Pairing in DNA

The two strands of a DNA molecule are **complementary** to each other which means that the sequence of each strand determines the sequence of the other.

An A on one strand means a T on the opposite strand, and a G on one strand means a C on the other which allows us to predict the sequence of one strand by just looking at the complementary strand.

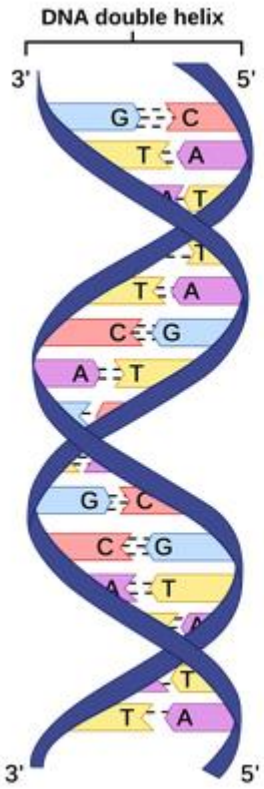
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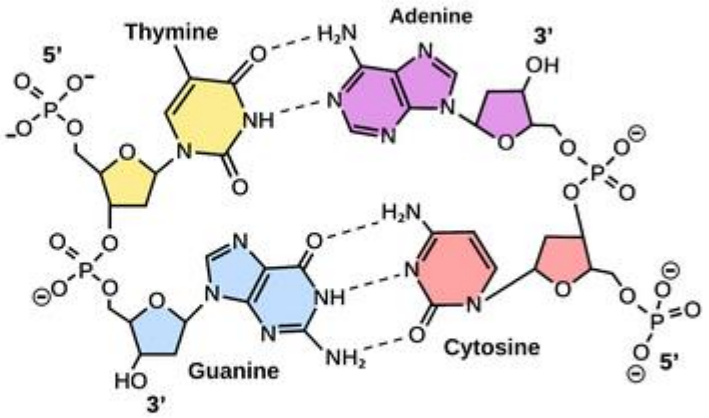
The structure of DNA suggests the basic mechanism of its replication.



DNA sequence

- The base pairing rules do not restrict the sequence of nucleotides along each DNA strand.
- The linear sequence of the four bases can be varied in countless ways.
- Each gene has a specific sequence of nucleotide bases.



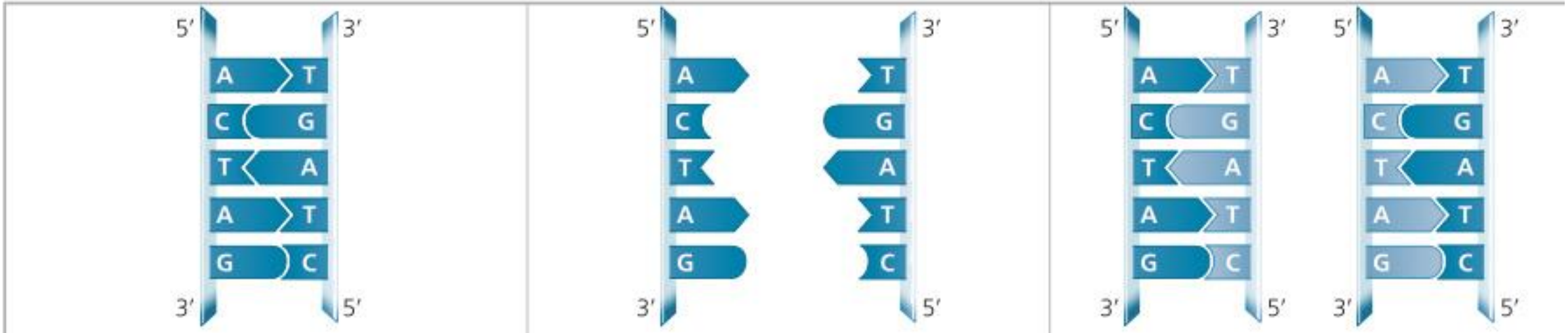


The Basic Principle of DNA replication: Base Pairing to a Template Strand

Watson and Crick noted that the specific base pairing suggested a possible copying mechanism for DNA.

Since the two strands of DNA are complementary, each strand acts as a template for building a new strand in replication (copying of DNA).

Prior to DNA replication the hydrogen bonds are broken, and the two chains unwind and separate and two new daughter strands are built based on base-pairing rules.



(a) The parental molecule has two complementary strands of DNA. Each base is paired by hydrogen bonding with its specific partner, A with T and G with C.

(b) First, the two DNA strands are separated. Each parental strand can now serve as a template for a new, complementary strand.

(c) Nucleotides complementary to the parental (dark blue) strand are connected to form the sugar-phosphate backbones of the new "daughter" (light blue) strands.

DNA replication is semiconservative

The Basic Principle of DNA replication: Base Pairing to a Template Strand

Watson and Crick's **semiconservative model of replication** predicts that when a DNA double helix replicates, each daughter molecule will have one old strand derived from the parent molecule and one newly made strand.

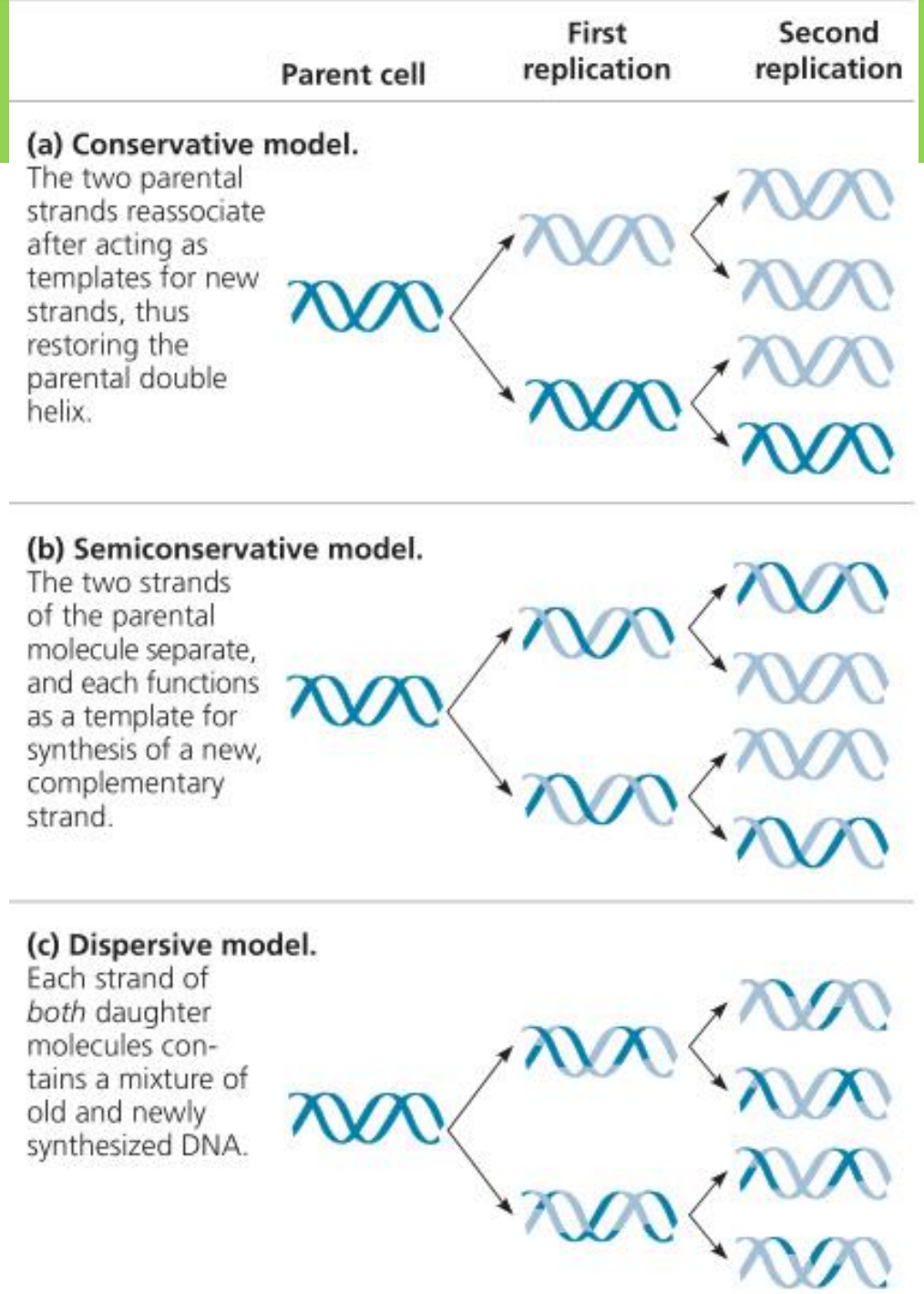
• During DNA replication, base pairing enables existing DNA strands to serve as templates for new complementary strands.

– The order of bases on one strand can be used to add complementary bases and therefore duplicate the pairs of bases exactly.

The complementary nucleotides line up and are connected to form the sugar-phosphate backbones of the new strands.

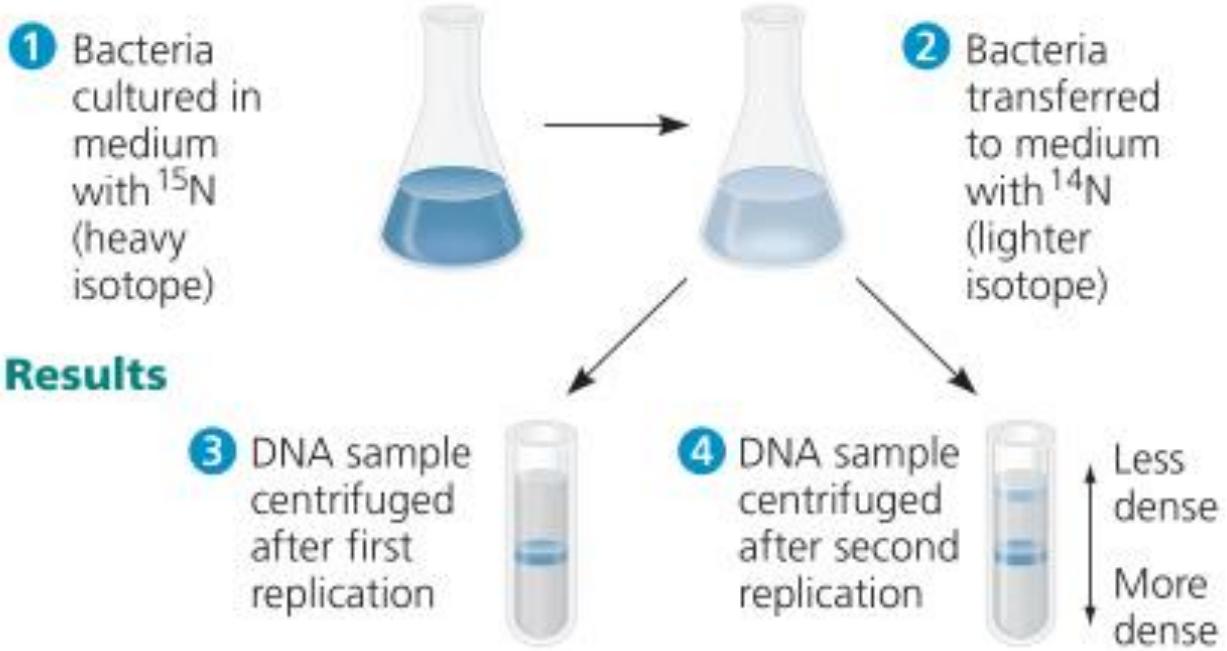
Alternative Models of Replication

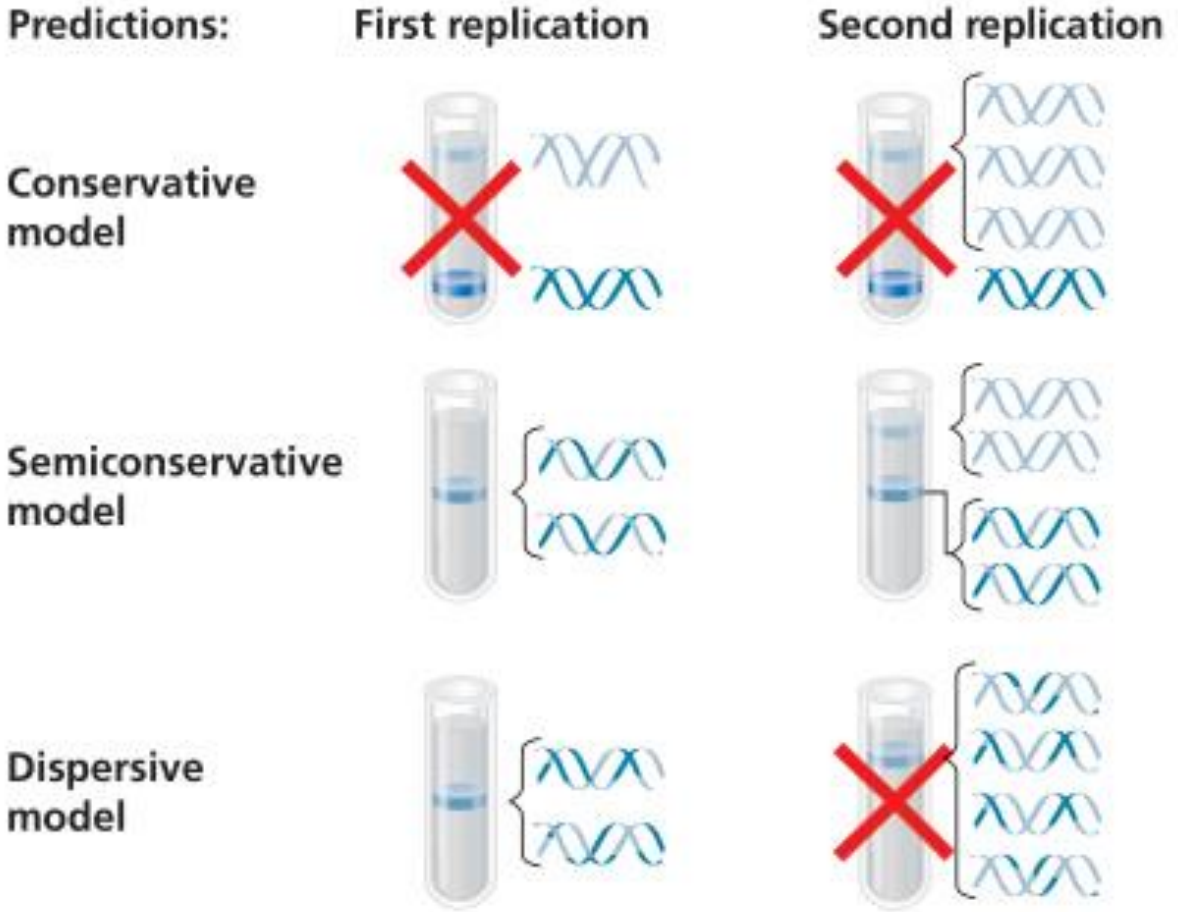
- Experiments by Matthew Meselson and Franklin Stahl supported the semiconservative model
- They labeled the nucleotides of the old strands with a heavy isotope of nitrogen, while any new nucleotides were labeled with a lighter isotope The first replication produced a band of hybrid DNA, eliminating the conservative model
- A second replication produced both light and hybrid DNA, eliminating the dispersive model and supporting the semiconservative model



Meselson and Stahl's Experiment

Experiment Matthew Meselson and Franklin Stahl cultured *E. coli* for several generations in a medium containing nucleotide precursors labeled with a heavy isotope of nitrogen, ^{15}N. They then transferred the bacteria to a medium with only ^{14}N, a lighter isotope. They took one sample after the first DNA replication and another after the second replication. They extracted DNA from the bacteria in the samples and then centrifuged each DNA sample to separate DNA of different densities.





DNA Replication: A Closer Look

- E. Coli has 4.6 million base pairs in its single chromosome and it takes less than an hour to copy its entire chromosome and divide to form two identical daughter cells.

- A human somatic cell has 6 billion base pairs. It takes a few hours to replicate the DNA during S phase.

- DNA replication is fast and remarkably accurate, with only 1 error per 10 billion nucleotides.

- More than a dozen enzymes and other proteins participate in DNA replication.

DNA Replication: Getting Started

• Replication begins at special sites called **origins of replication**, where the two DNA strands are separated, opening up a **replication "bubble".**

At each end of a replication bubble is a **replication fork**, a Y-shaped region where the parental strands of DNA are being unwound.

• Enzymes are responsible for the separation and unwinding of strands.

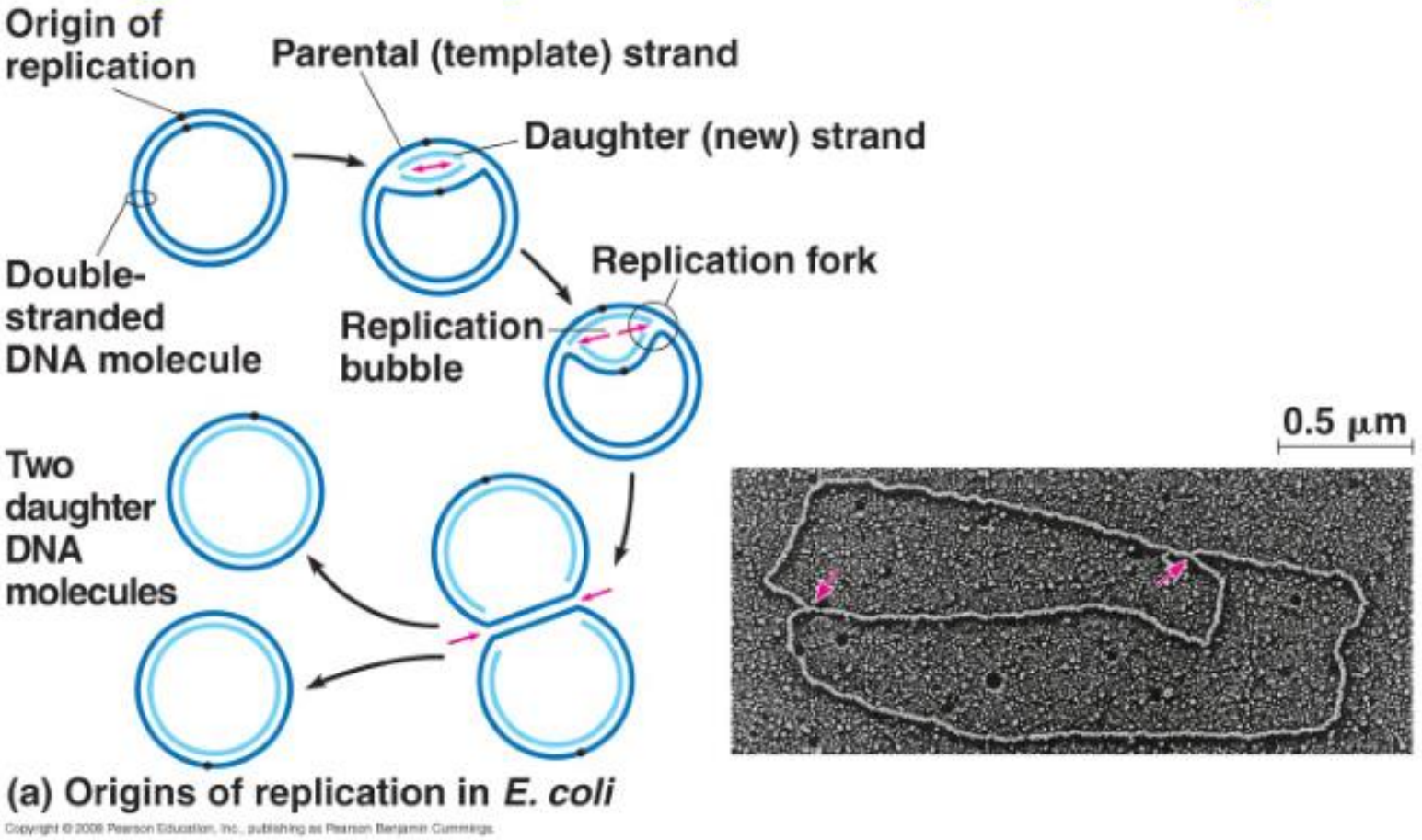
• Replication proceeds in both directions from each origin, until the entire molecule is copied.

• In eukaryotes, there may be hundreds or thousands of origin sites per chromosome.

• Replication bubbles have replication forks at each end, where the parental strands of DNA are being unwound.

• The replication bubbles elongate as the DNA is replicated, and eventually fuse.

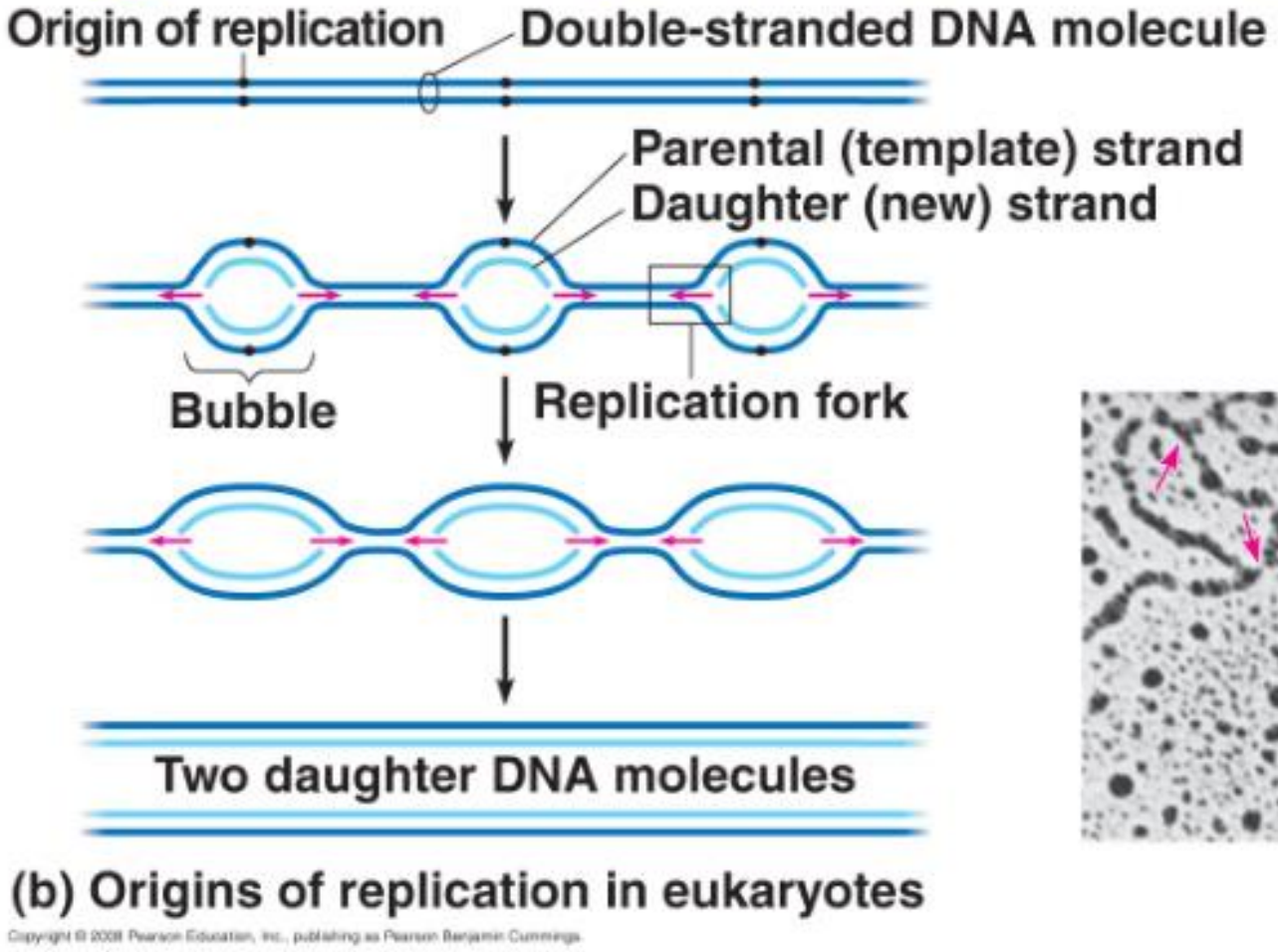
Origin of Replication in Prokaryote



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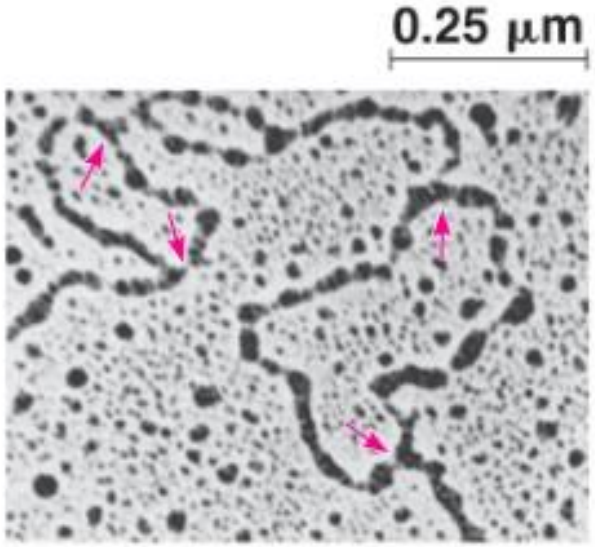
Many bacteria have circular chromosomes with only one origin of replication.
The parental strands separate at the origin and form a replication bubble with two forks.
Replication proceeds in both directions until the forks meet.
This results in 2 daughter DNA molecules.

Origins of Replication in Eukaryotes



(b) Origins of replication in eukaryotes

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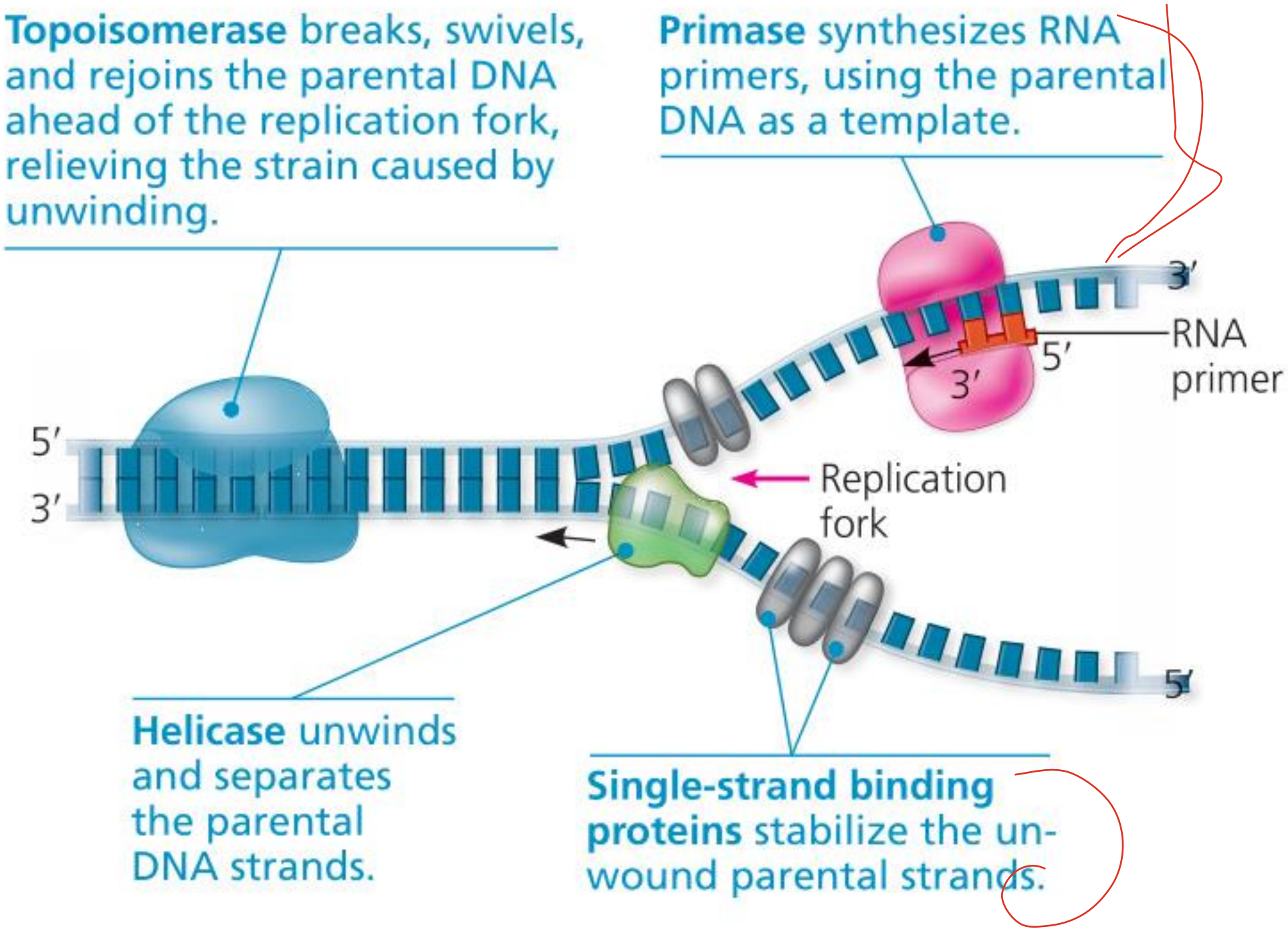


Eukaryotes have linear chromosomes.

DNA replication begins when replication bubbles form at many sites along the DNA molecule.

The bubbles expand as replication proceeds in both directions.

Eventually the bubbles fuse and synthesis of the daughter DNA strands is complete.


Topoisomerase breaks, swivels, and rejoins the parental DNA ahead of the replication fork, relieving the strain caused by unwinding.
Primase synthesizes RNA primers, using the parental DNA as a template.
3′
RNA primer
5′
3′
5′
3′
Replication fork
5′
Helicase unwinds and separates the parental DNA strands.
Single-strand binding proteins stabilize the unwound parental strands.

Proteins involved in DNA unwinding

- **Topoisomerases** unwind the DNA helix, giving access to helicases (unzipping enzymes) to bind to the double stranded DNA at the origin. They correct —overwinding ahead of replication forks by breaking, swiveling, and rejoining DNA strands.

- **Helicases** break the hydrogen bonds holding the two strands together, resulting in two separate strands.

- **Single-strand binding proteins** keep the unpaired template strands apart during replication.

- The parental DNA strands are now available to serve as templates for the synthesis of new complementary DNA strands.

Synthesizing a New DNA Strand

• Enzymes that catalyze DNA synthesis are called **DNA polymerases**.

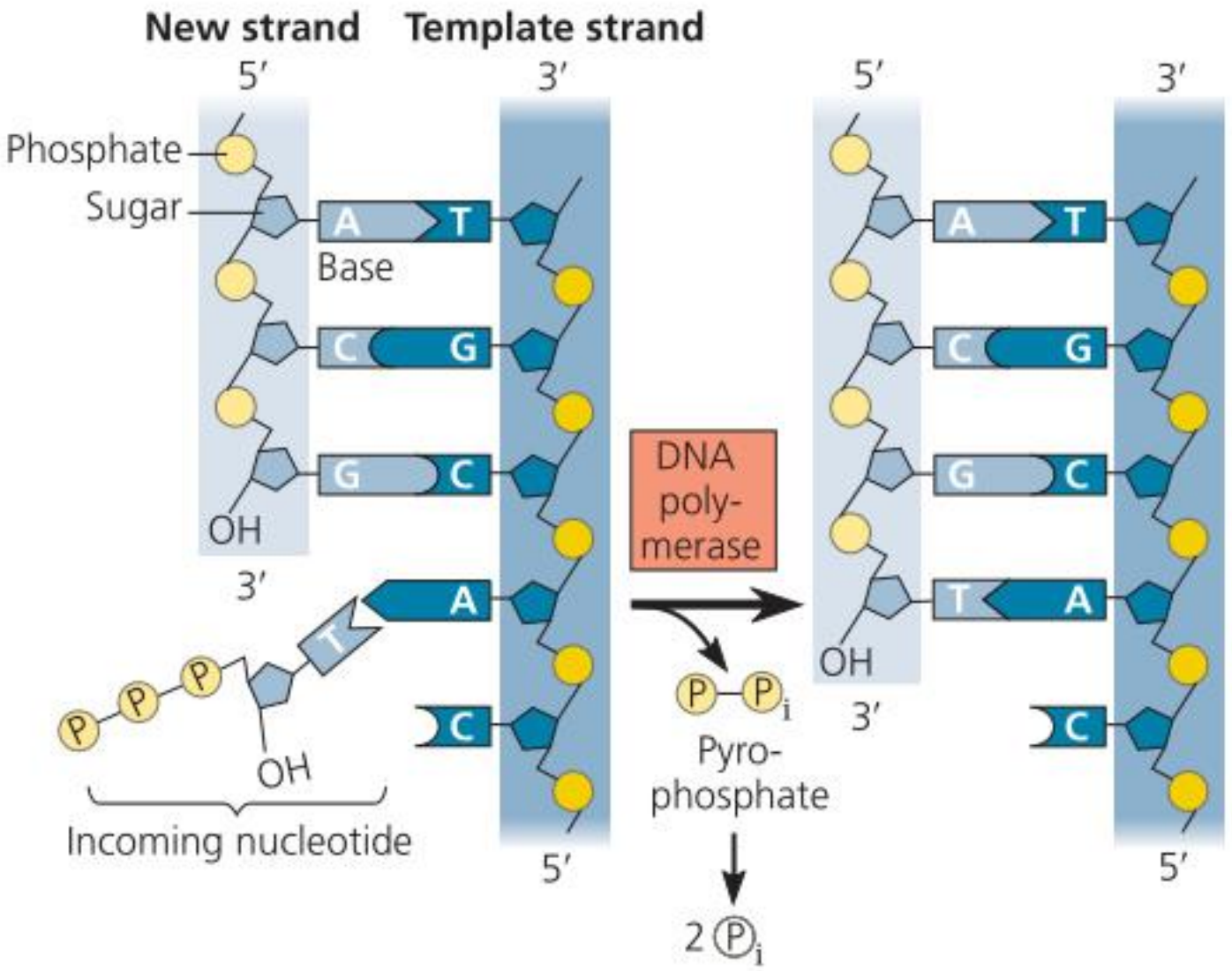
DNA polymerases cannot initiate synthesis of DNA from scratch; they can only add nucleotides to the 3’ end of a preexisting chain.

• The initial nucleotide strand which DNA polymerase adds nucleotides to is a short RNA strand, called **primer**.

The primer is short (5–10 nucleotides long), and the 3’ end serves as the starting point for the new DNA strand.

An enzyme called **primase** can start an RNA chain from scratch and adds RNA nucleotides one at a time using the parental DNA as a template.

After the synthesis of primer, **DNA polymerase III** catalyzes the elongation of new DNA at a replication fork using the parental DNA strand as a template.



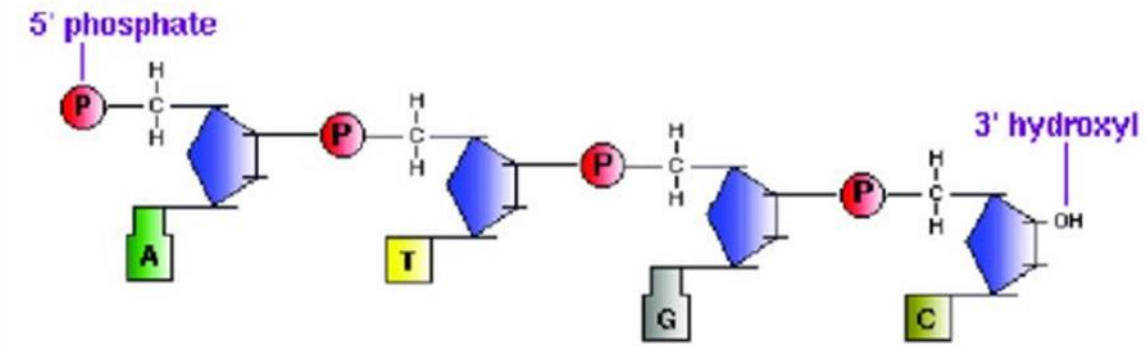
DNA polymerase III catalyzes the addition of each monomer to the growing end of a DNA strand by a condensation reaction in which two phosphate groups are lost as a molecule of pyrophosphate.

Subsequent hydrolysis of the pyrophosphate to two molecules of inorganic phosphate is a coupled that helps drive the polymerization reaction.

The rate of elongation of bacterial DNA = 500 nucleotides per second.

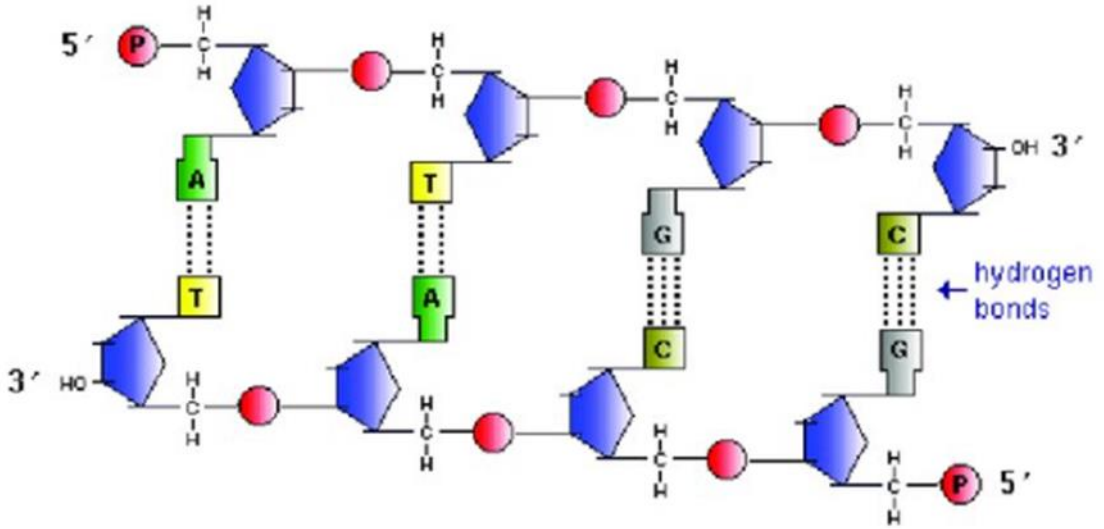
The rate of elongation of human DNA = 50 nucleotides per second in human cells.

Antiparallel Elongation



- The sugar-phosphate backbones of DNA are antiparallel: they run in opposite directions, like one- way streets.

- Each DNA strand has a 3′ end with a free hydroxyl group (OH) attached to deoxyribose and a 5′ end with a free phosphate group attached to deoxyribose.

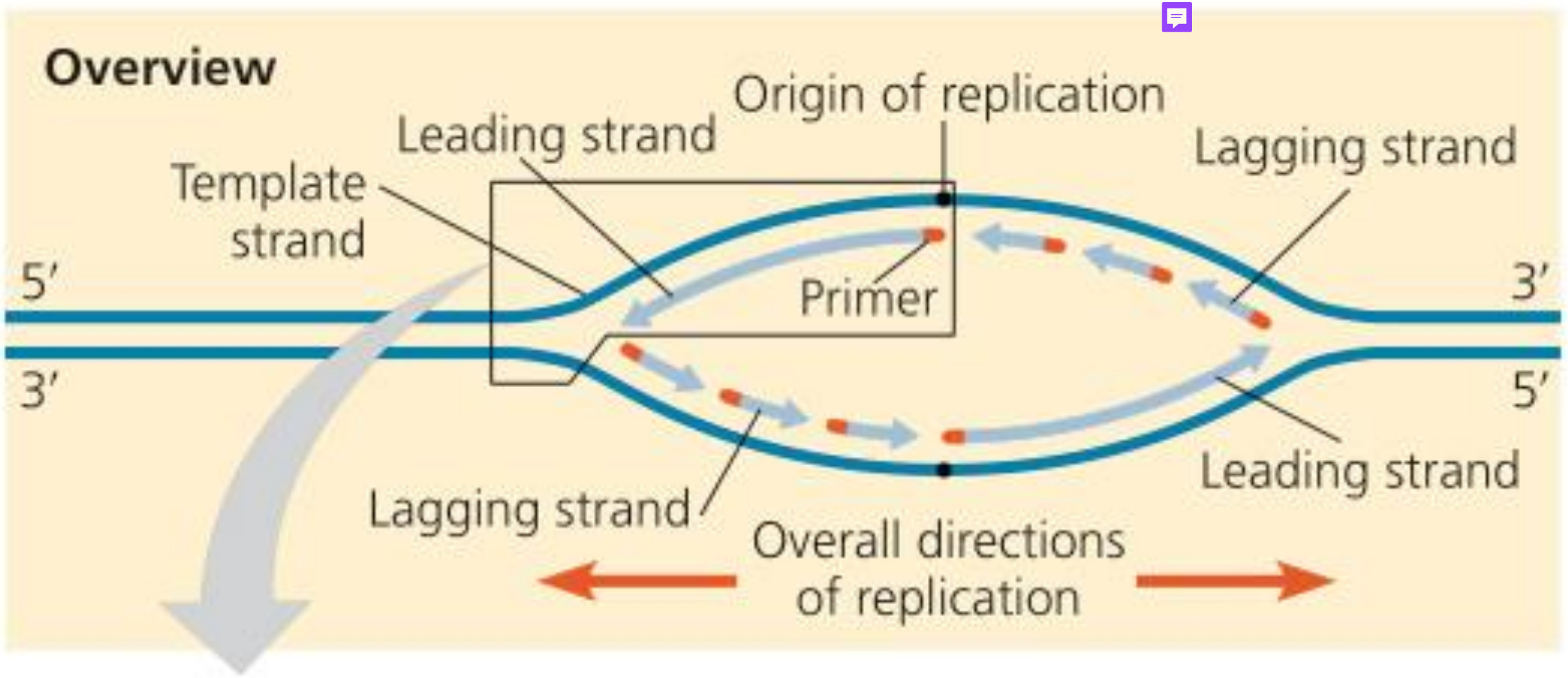


- The 5′->3′ direction of one strand runs opposite the 3′->5′ direction of the other strand.

- DNA polymerases can add nucleotides only to the free 3′ end of a growing DNA strand.

- A new DNA strand can elongate only in the 5′->3′ direction.

Antiparallel Elongation



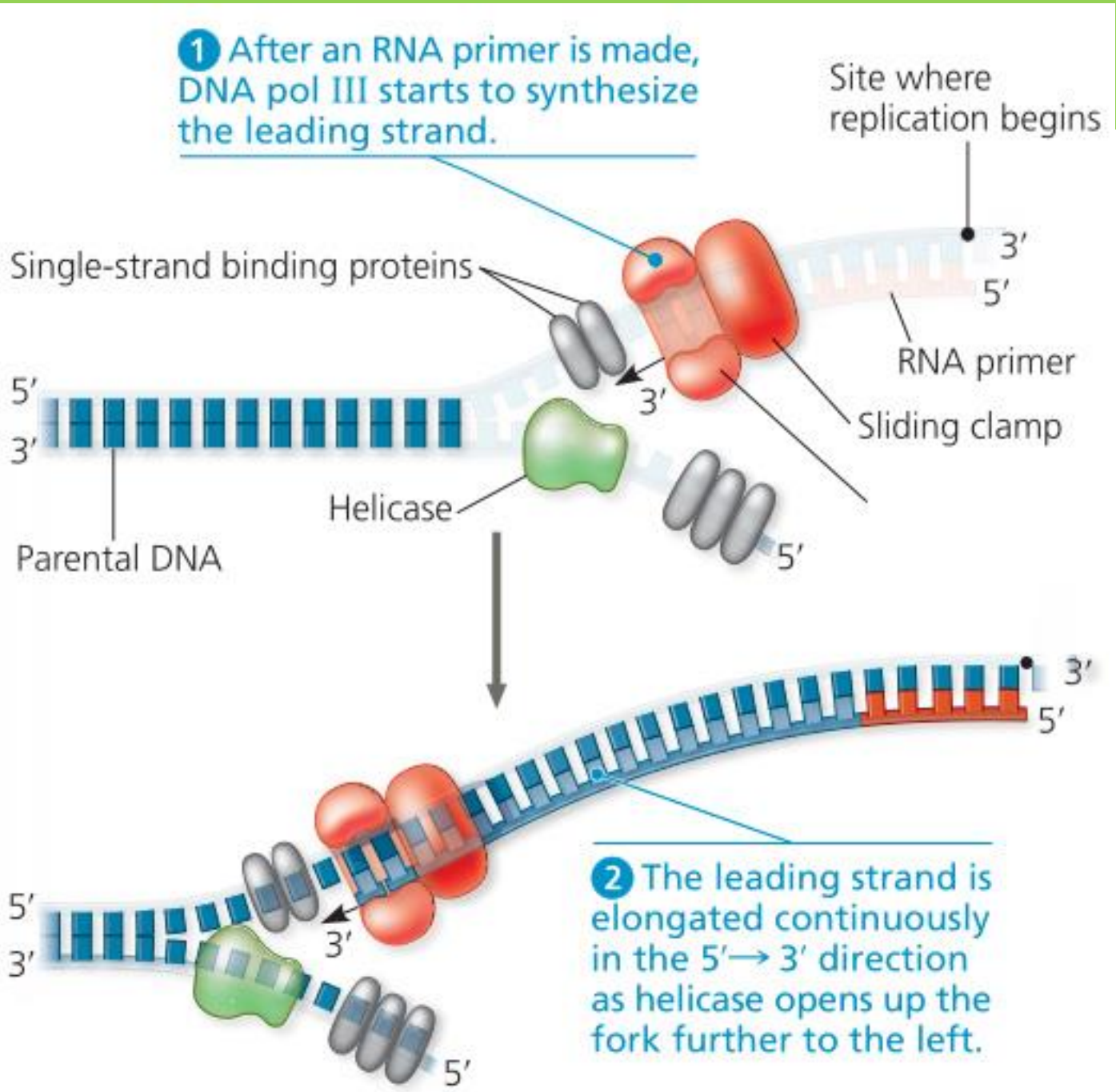
Antiparallel Elongation

Steps of leading strand in the synthesis of new DNA in the 5′->3′ direction

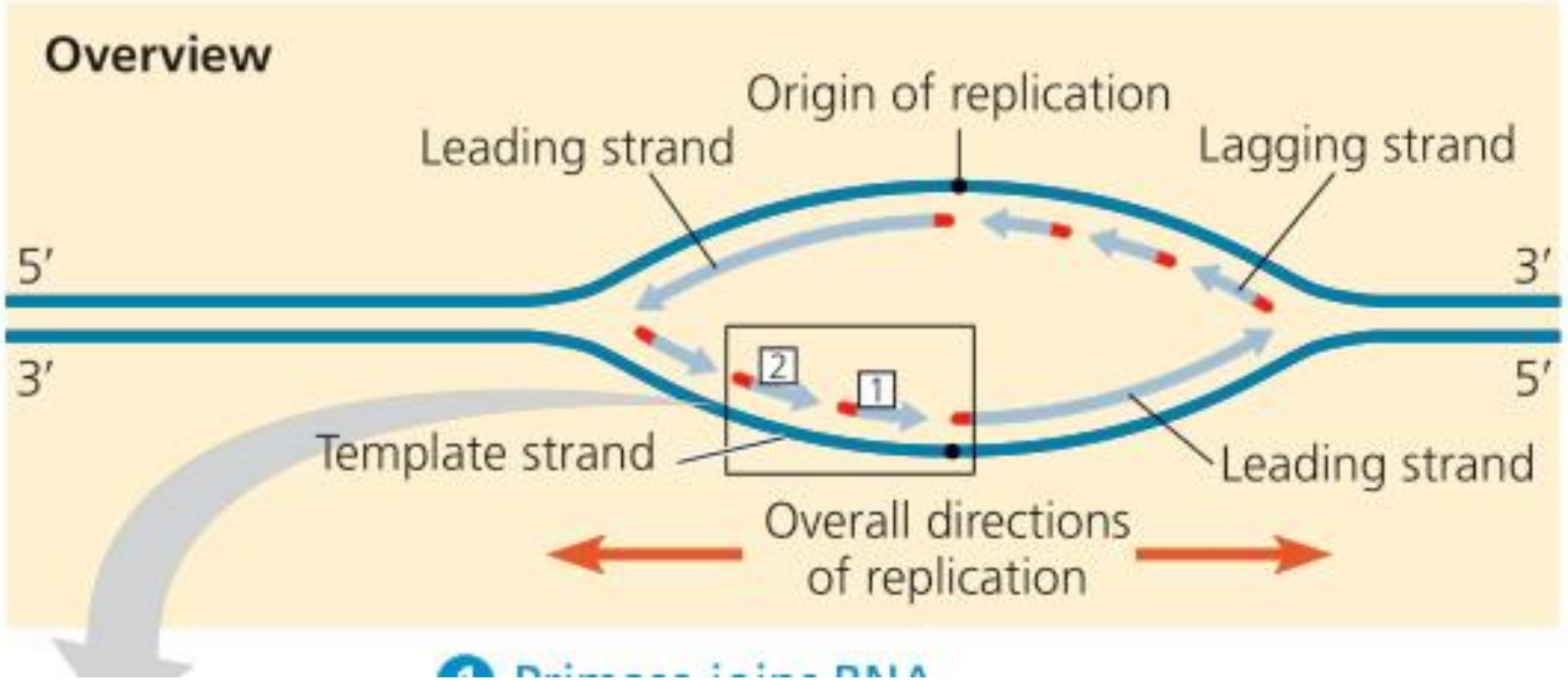
1.The DNA strands have been separated by Helicase at the replication fork.

2. A short RNA primer is assembled by primase.

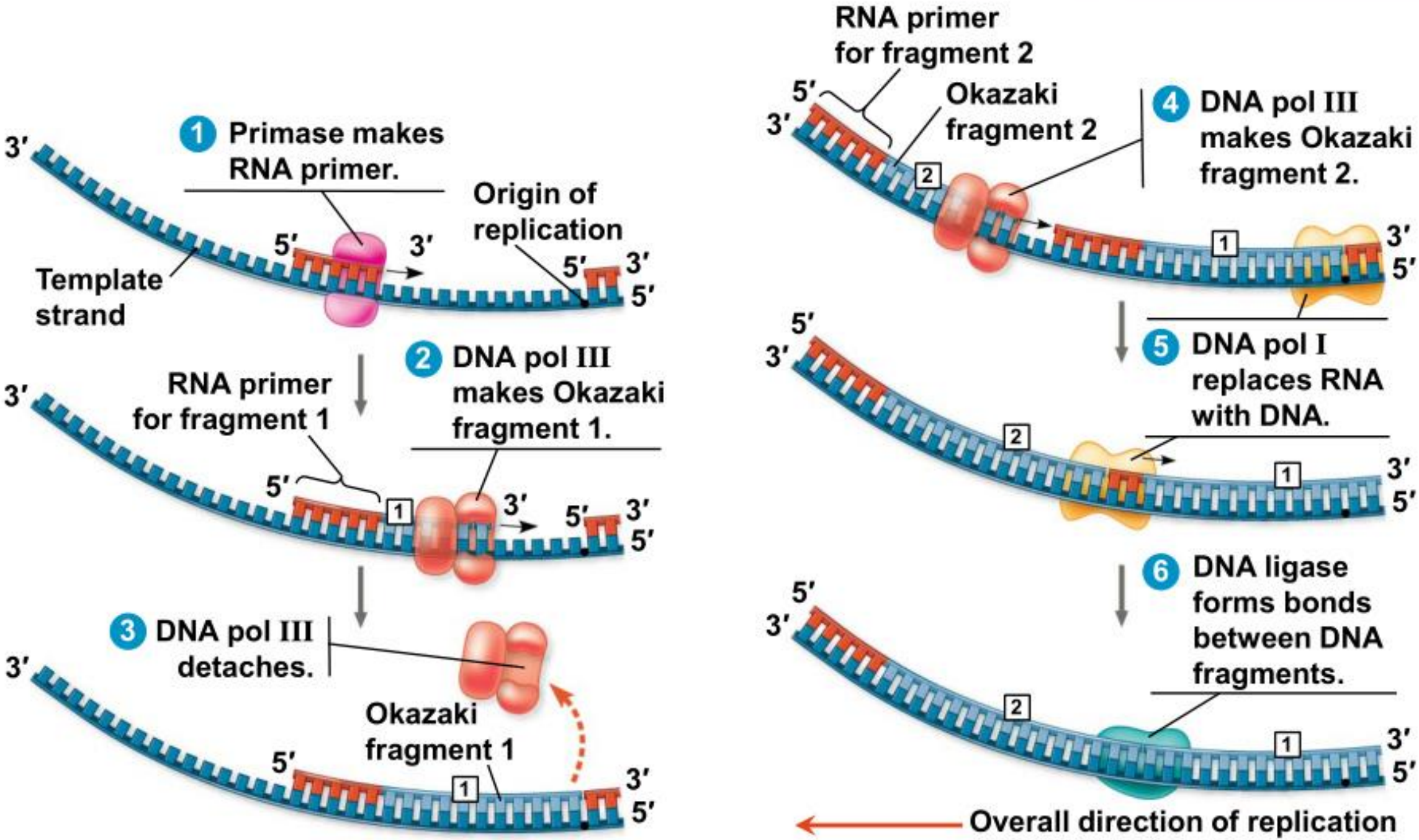
3. DNA polymerase III nestles in the replication fork (via the “sliding clamp”) and it can continuously add nucleotides to the new DNA strand in the 5->3′ direction.

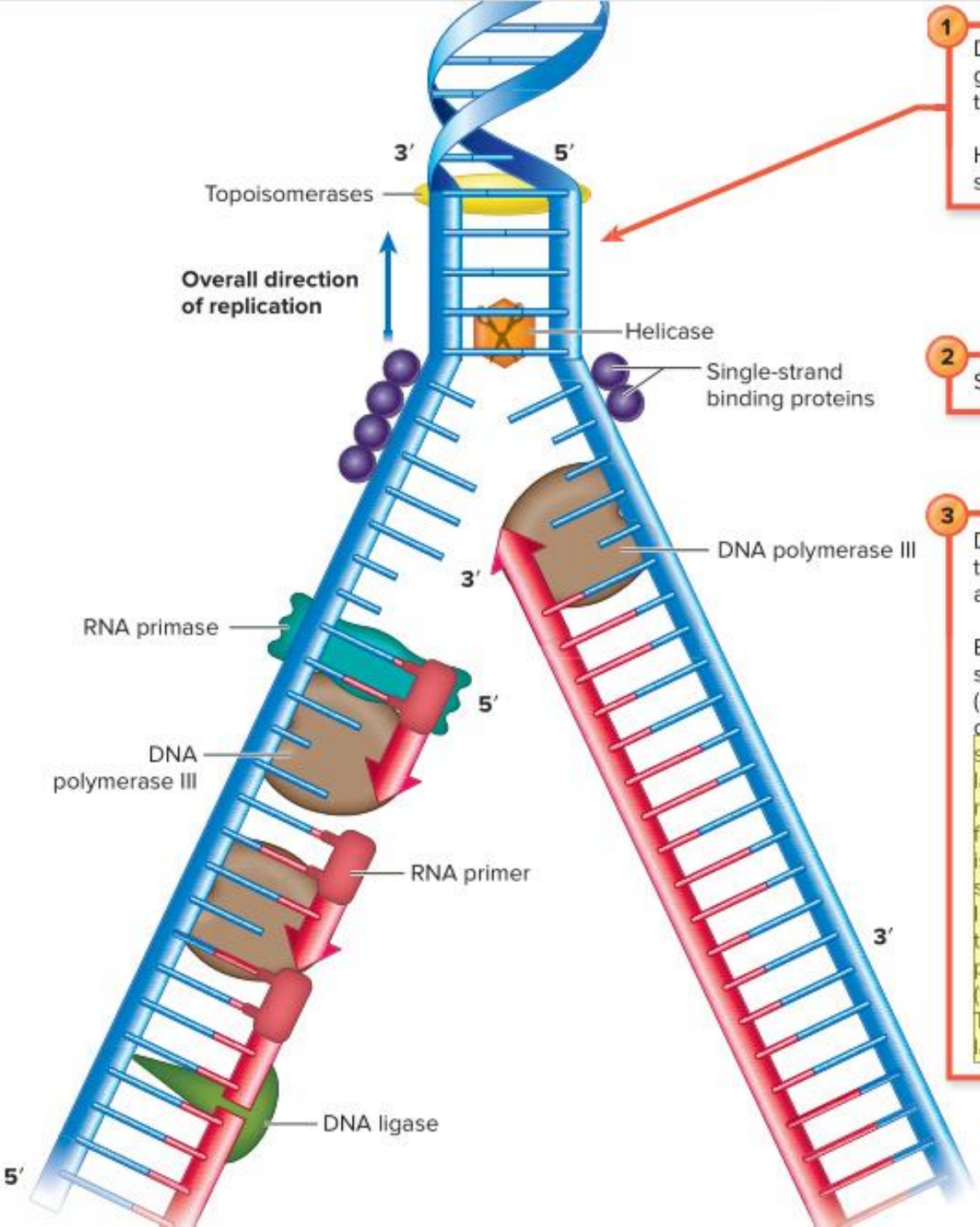


▼ **Figure 16.17 Synthesis of the lagging strand.**



- To elongate the other parental strand (5′->3′ into the fork), the lagging strand, DNA polymerase III must work in the direction away from the replication fork.
- The lagging strand is synthesized as a series of segments called **Okazaki fragments**, which are joined together by **DNA ligase**.
- Okazaki fragments are about 1,000 to 2,000 nucleotides long in E. coli and 100 to 200 nucleotides long in eukaryotes.


1 Primase makes RNA primer.
Origin of replication
Template strand
2 DNA pol III makes Okazaki fragment 1.
RNA primer for fragment 1
3 DNA pol III detaches.
Okazaki fragment 1
RNA primer for fragment 2
Okazaki fragment 2
4 DNA pol III makes Okazaki fragment 2.
5 DNA pol I replaces RNA with DNA.
6 DNA ligase forms bonds between DNA fragments.
Overall direction of replication
3′
5′
1
2



1 During replication topoisomerases unwind the DNA helix, giving access to helicases (unzipping enzymes) to bind to the dsDNA at the origin.

Helicases break the hydrogen bonds holding the two strands together, resulting in two separate strands.

2 Single-stranded binding proteins keep the strands apart.

3 DNA polymerase III adds nucleotides in accordance with the template pattern. Note that RNA primase will have already added a short length of RNA.

Because DNA polymerase is correctly oriented for synthesis only in the 5′ to 3′ direction of the new molecule (red) strand, only one strand, called the leading strand, can be synthesized as a continuous, complete strand. The strand with the opposite orientation (3′ to 5′) is termed the lagging strand. On this strand the polymerase adds nucleotides a few at a time in the direction away from the fork (5′ to 3′). As the fork opens up a bit, the next segment is synthesized backward to the point of the previous segment, a process repeated until synthesis is complete. In this way, the DNA polymerase is able to synthesize the two new strands simultaneously. This manner of synthesis produces one strand containing short fragments of DNA (100 to 1,000 bases long) called Okazaki fragments. These fragments are attached to the growing end of the lagging strand by another enzyme called DNA ligase.

Figure 6.6 DNA replication

Antiparallel Elongation

Synthesis of the Lagging Strand in the 3′->5′ direction
The DNA strands have been separated by Helicase at the replication fork.
1. A short RNA primer is made by primase.
2. DNA polymerase III adds nucleotides to the new DNA strand in the 3′->5′ direction to form Okazaki fragment #1.
3. After reaching the next RNA primer to the right, DNA ploymerase III detatches.
4. After Okazaki fragment #2 is primed (by primase) DNA ploymerase III will attach and add nucleotides to this fragment in the 3′→5′ direction until it reaches the Okazaki fragment #1primer, then DNA polymerase III detaches.
5. This process repeats, creating several Okazaki fragments along the lagging strand.

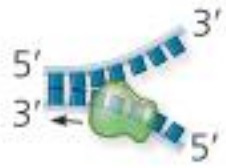
Antiparallel Elongation

DNA polymerase I replaces the RNA on the 5’ side of Okazaki fragment #1 with DNA adding it to the 3’ end of fragment #2.

DNA polymerase I cannot join the final nucleotide of the replacement DNA segment to the DNA of the Okazaki fragment 1. After the last addition, the backbone is left with a free 3’ end.

- **DNA ligase** joins the sugar-phosphate backbones of all the Okazaki fragments into a continuous DNA strand.

Bacterial DNA Replication Proteins and Their Functions

Protein	Function
Helicase 	Unwinds parental double helix at replication forks
Single-strand binding protein 	Binds to and stabilizes single-stranded DNA until it is used as a template
Topoisomerase 	Relieves overwinding strain ahead of replication forks by breaking, swiveling, and rejoining DNA strands
Primase 	Synthesizes an RNA primer at 5′ end of leading strand and at 5′ end of each Okazaki fragment of lagging strand
DNA pol III 	Using parental DNA as a template, synthesizes new DNA strand by adding nucleotides to an RNA primer or a pre-existing DNA strand
DNA pol I 	Removes RNA nucleotides of primer from 5′ end and replaces them with DNA nucleotides added to 3′ end of adjacent fragment
DNA ligase 	Joins Okazaki fragments of lagging strand; on leading strand, joins 3′ end of DNA that replaces primer to rest of leading strand DNA

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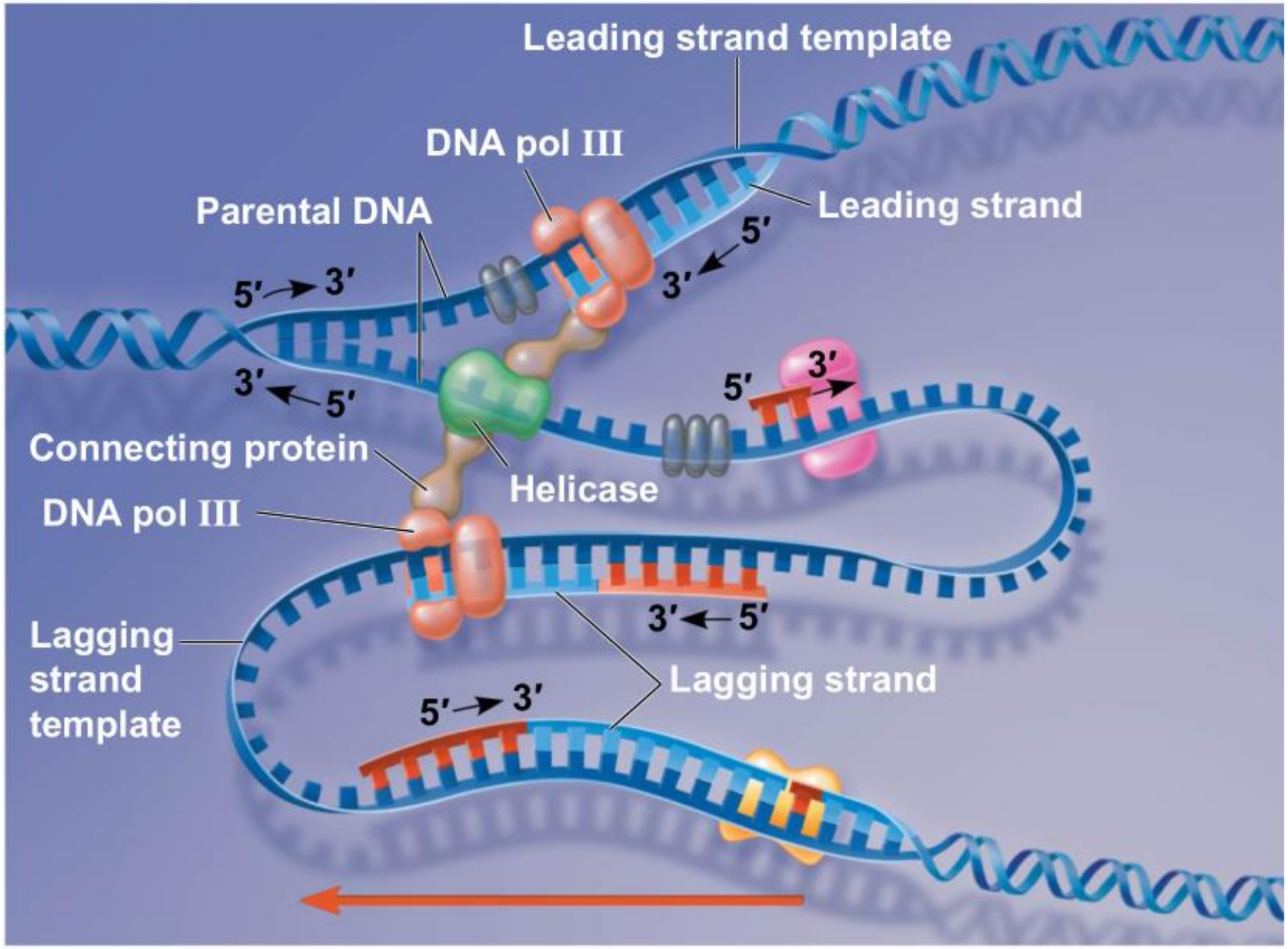
The DNA Replication Complex

• The proteins that participate in DNA replication form a large complex, a “DNA replication machine”

• the DNA replication complex may not move along the DNA; rather, the DNA may move through the complex during the replication process.

In eukaryotic cells, multiple copies of the complex, perhaps grouped into “factories,” may be anchored to the nuclear matrix.

• Recent studies support a model in which two DNA polymerase molecules, one on each template strand, “reel in” the parental DNA and extrude newly made daughter DNA molecules. In this so-called trombone model, the lagging strand is also looped back through the complex.



The “trombone” model of the DNA replication complex. In this proposed model, two molecules of DNA polymerase III work together in a complex, one on each strand, with helicase and other proteins. The lagging strand template DNA loops through the complex, resembling the slide of a trombone.



Proofreading and Repairing DNA

Mistakes during the initial pairing of template nucleotides and complementary nucleotides occur at a rate of one error per 100,000 base pairs.

DNA polymerase proofreads each new nucleotide against the template nucleotide as soon as it is added and replaces incorrect nucleotides.

– The final error rate is only one mistake per 10 billion nucleotides.

(Human DNA consists of 3 billion nucleotides)

A **mismatched nucleotide** sometimes evade proofreading by a DNA polymerase.

In **mismatch repair**, other enzymes remove and replace incorrectly paired nucleotides that have resulted from replication errors.

Because repair of damaged DNA is so important to the survival of an organism, it is no surprise that many different DNA repair enzymes have evolved. Almost 100 are known in E. coli, and about 170 have been identified so far in humans.

A hereditary defect in one of these enzymes is associated with a form of colon cancer.

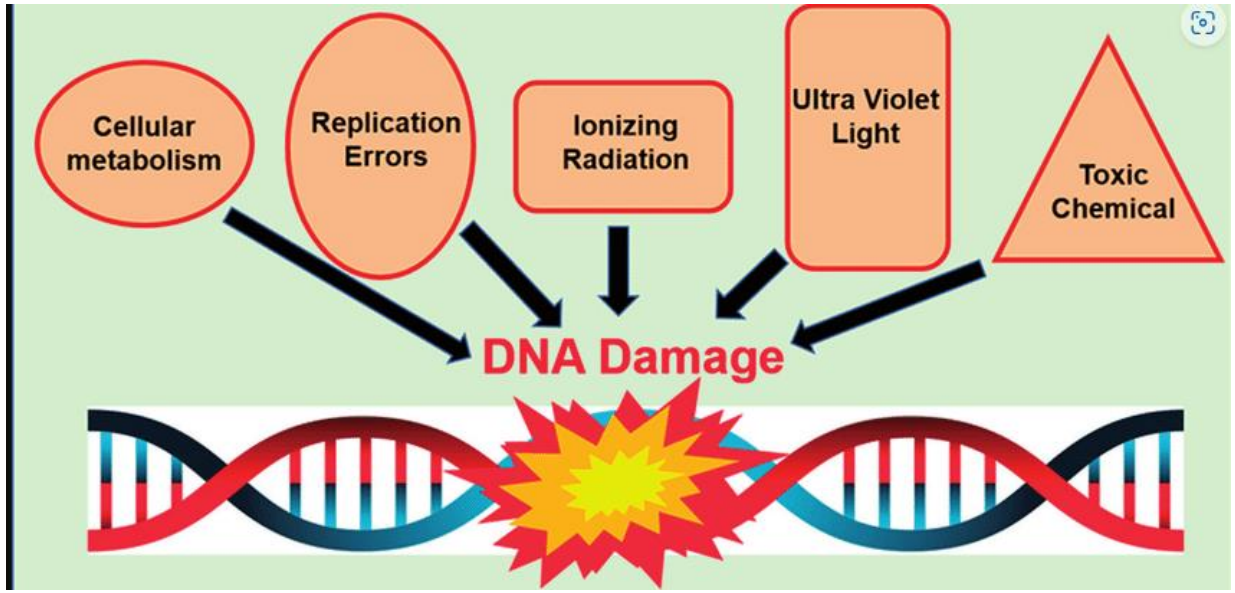
DNA Damage Can Arise After Replication

Incorrectly paired or altered nucleotides can also arise after replication.

- DNA molecules are constantly subjected to potentially harmful chemical and physical agents.
- Reactive chemicals, radioactive emissions, X-rays, ultraviolet light, and molecules in cigarette smoke can change nucleotides in ways that can affect encoded genetic information.
- Furthermore, DNA bases may undergo spontaneous chemical changes under normal cellular conditions.

However, these changes in DNA are usually corrected before they become permanent changes (mutations).

Each cell continuously monitors and repairs its genetic material.



Nucleotide Excision Repair of DNA

- Many cellular systems for repairing incorrectly paired nucleotides use a mechanism that relies on the base-paired structure of DNA.

- A **nuclease** (a DNA-cutting enzyme) cuts out a segment of a damaged strand, and the resulting gap is filled in with nucleotides, using the undamaged strand as a template.

- One such DNA repair system is called **nucleotide excision repair**.

- DNA repair enzymes in skin cells repair genetic damage caused by ultraviolet light.

- Ultraviolet light can produce thymine dimers between adjacent thymine nucleotides.

- This buckles the DNA double helix and interferes with DNA replication.

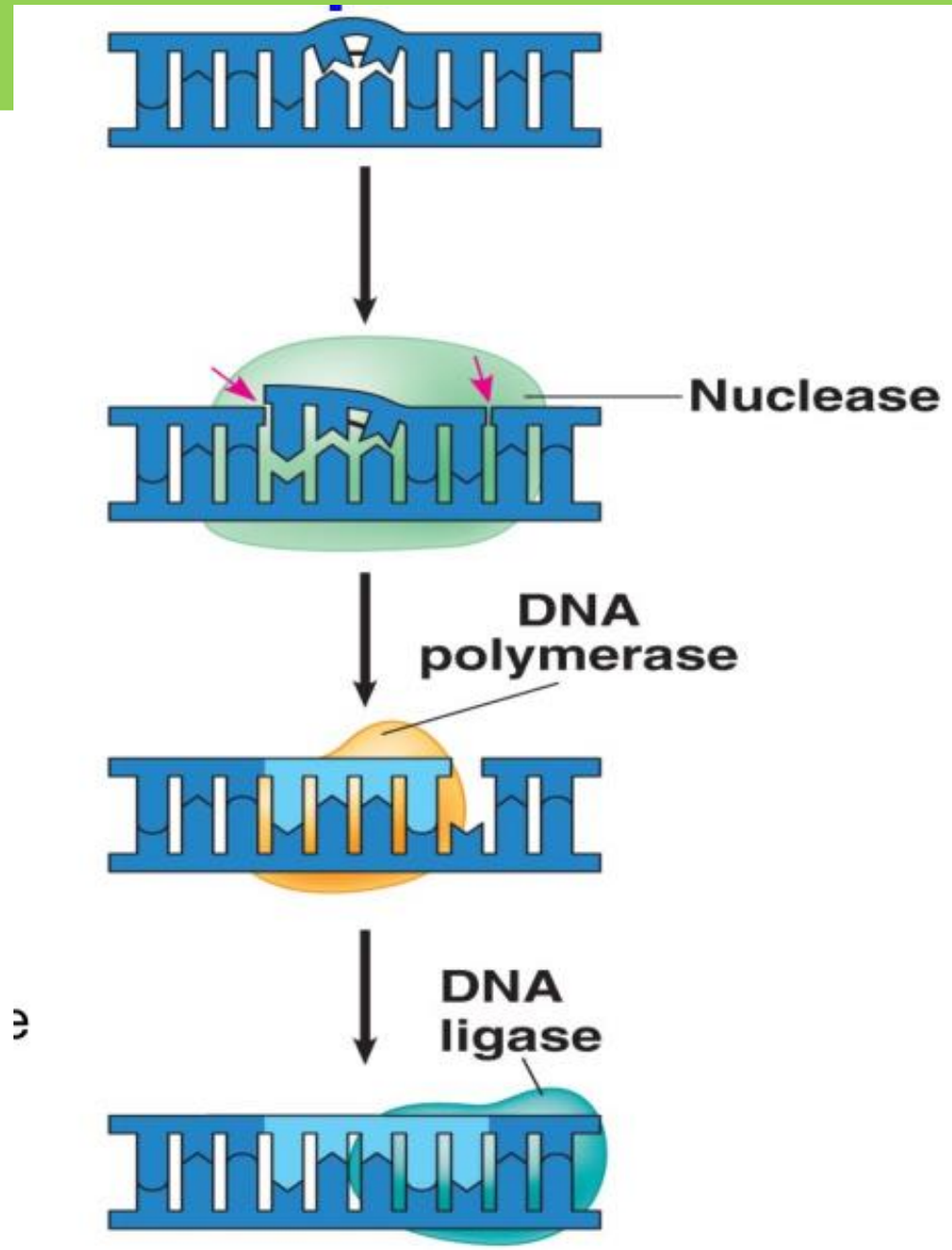
Nucleotide Excision Repair of DNA

1. A thymine dimer causes a buckle in the DNA strand.

2. A nuclease cuts the damaged DNA at 2 points and the damaged section is removed.

3. A DNA polymerase fill in the missing nucleotides complimentary to the undamaged strand.

4. DNA ligase combines the free end of the new DNA with the old DNA.



Evolutionary Significance of Altered DNA Nucleotides

- Error rate after proofreading repair is low but not zero.

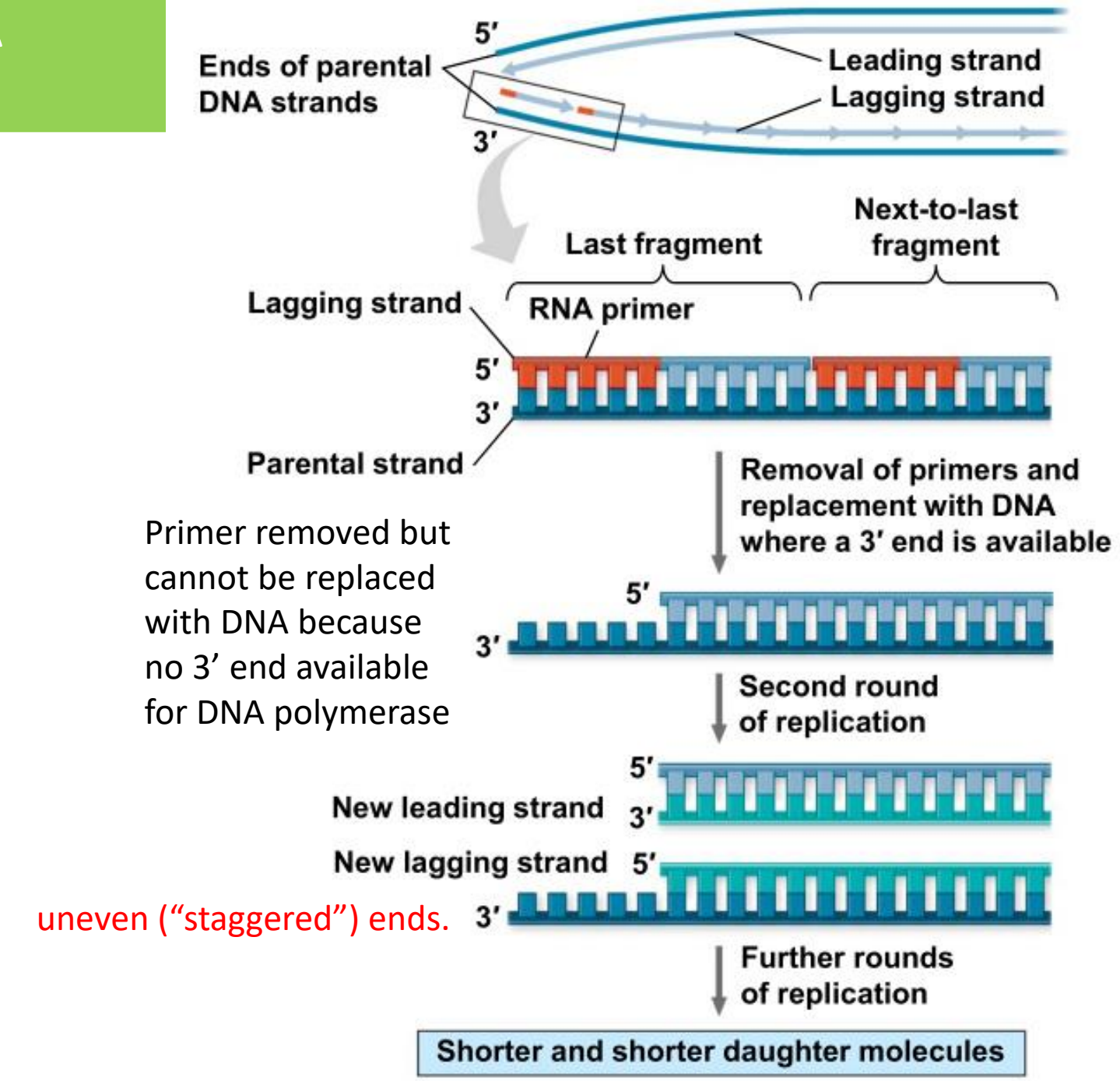
- Mutations (a permanent change in the DNA sequence) can change the phenotype of an organism.

If they occur in germ cells mutations can be passed on from generation to generation.

- Mutations are the source of the genetic variation upon which natural selection operates

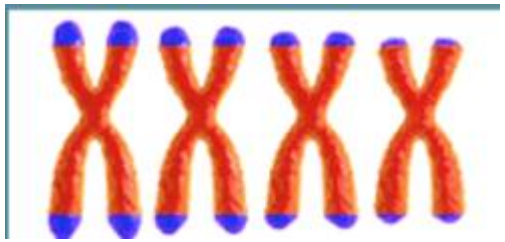
Replicating the Ends of DNA Molecules

– Prokaryotes do not have this problem because they have circular DNA molecules without ends.



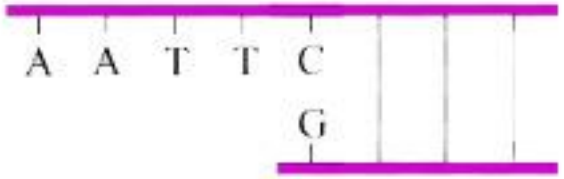
Replicating the Ends of DNA Molecules

• Eukaryotic chromosomal DNA molecules have special nucleotide sequences called **telomeres** at their ends.
Telomeres do not contain genes, instead they consist of multiple repetitions of one short nucleotide sequence.



For example, in each human telomere, the six-nucleotide sequence TTAGGG is repeated between 100 and 1,000 times.

Specific proteins associated with telomeric DNA prevent the staggered ends of the daughter molecule from activating the cell's systems for monitoring DNA damage. (Staggered ends of a DNA molecule, which often result from double-strand breaks, can trigger signal transduction pathways leading to cell cycle arrest or cell death.)



Telomeric DNA acts as a kind of buffer zone that provides some protection against the shortening of organism's genes.

Replicating the Ends of DNA Molecules

Telomeres become shorter during every round of replication.

Telomeric DNA tends to be shorter in dividing somatic cells of older individuals.

It is possible that the shortening of telomeres is somehow connected with the aging process of certain tissues and perhaps to aging in general.

Eukaryotic cells have evolved a mechanism to restore shortened telomeres in germ cells, which give rise to gametes.

- If the chromosomes of germ cells became shorter with every cell cycle, essential genes would eventually be lost.

- An enzyme called **telomerase** catalyzes the lengthening of telomeres in eukaryotic germ cells, restoring their original length.

Replicating the Ends of DNA Molecules

- Telomerase is not present in most cells of multicellular organisms.

- Therefore, the DNA of dividing somatic cells and cultured cells (cell that are grown in laboratory) tends to become shorter.
  – Telomere length may be a limiting factor in the life span of certain tissues and of the organism.

- The shortening of telomeres might protect cells from cancerous growth by limiting the number of cell divisions

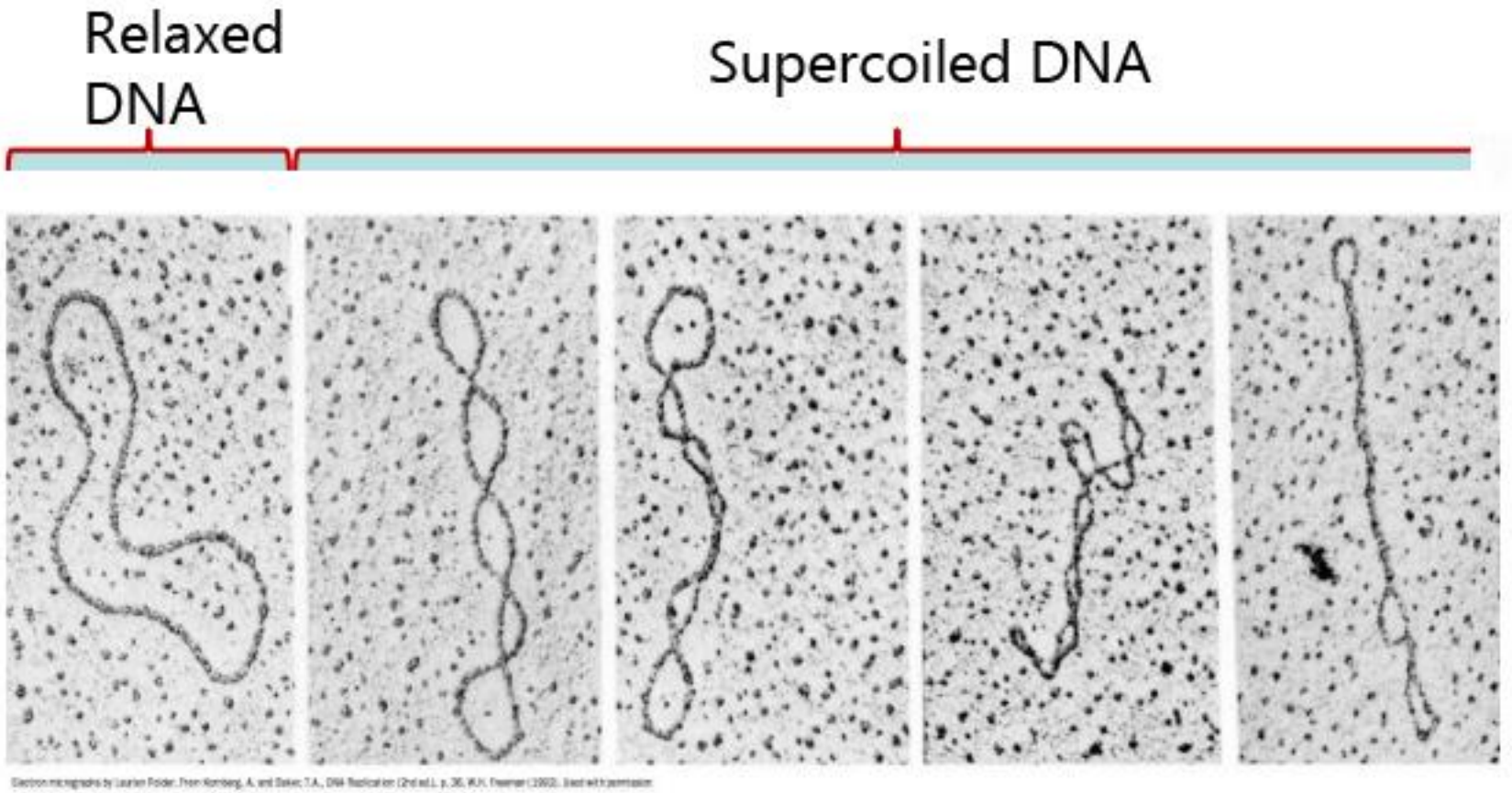
- There is evidence of telomerase activity in cancer cells, which may allow cancer cells to persist

A chromosome consists of a DNA molecule packed together with proteins

• Most bacteria have a single, circular, double-stranded DNA molecule, associated with a small amount of protein.

• The chromosomal DNA of E. coli consists of about 4.6 million nucleotide pairs, representing about 4,400 genes.

• In a bacterium, the DNA is "**supercoiled**" and found in a region of the cell called the **nucleoid** which is not bounded by membrane.



A chromosome consists of a DNA molecule packed together with proteins

A eukaryotic chromosome is a linear DNA molecule associated with a large amount of protein.

In eukaryotic cells DNA is packaged with protein to form chromatin

- **Chromatin**, a complex of DNA and protein, is found in the nucleus of eukaryotic cells

- Chromosomes fit into the nucleus through an elaborate, multilevel system of packing.

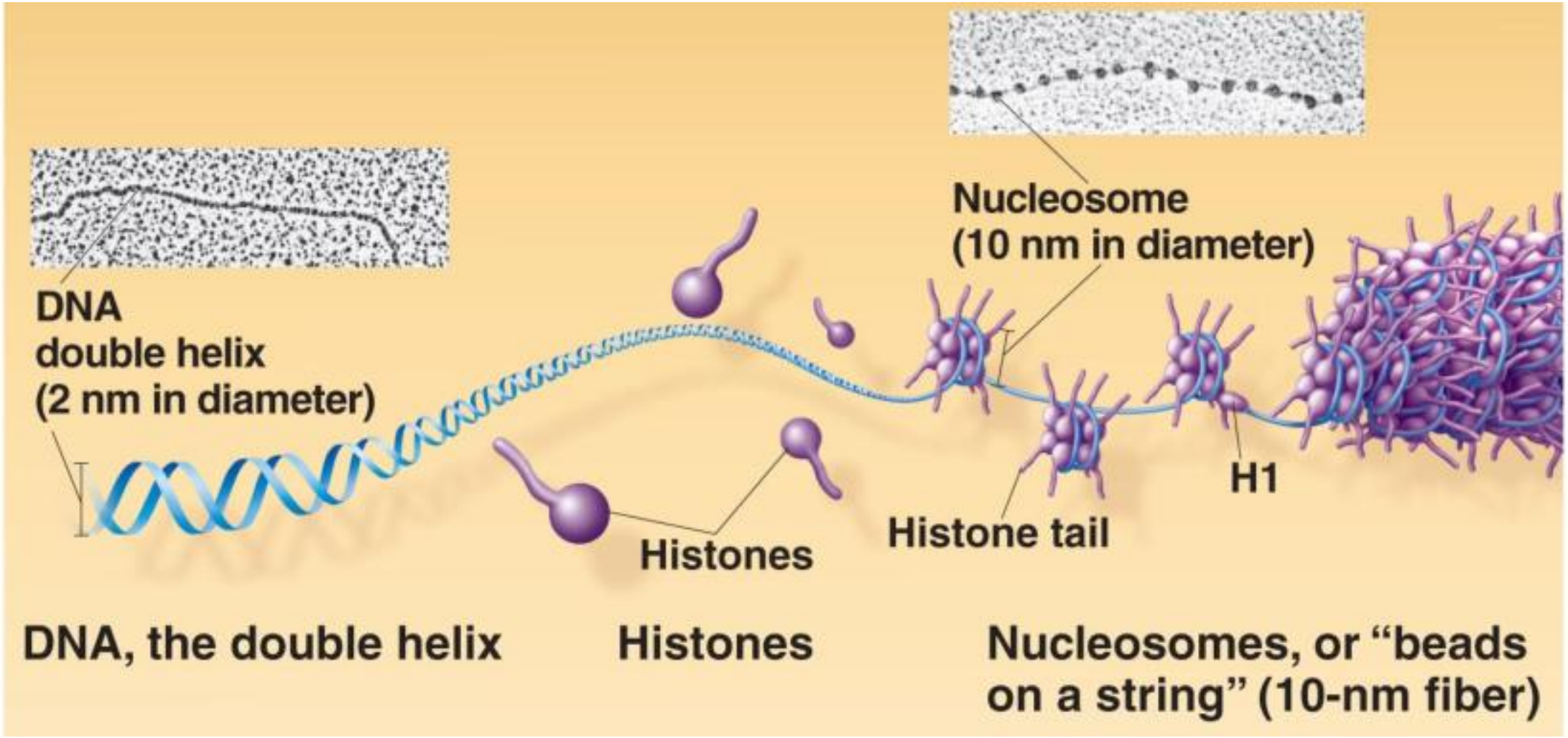
Chromatin is organized into fibers

– The packing of chromatin varies during the course of the cell cycle.

– It is highly extended during interphase.

– As a cell enters mitosis, the chromatin coils and condenses, forming short, thick metaphase chromosomes.

A chromosome consists of a DNA molecule packed together with proteins



A chromosome consists of a DNA molecule packed together with proteins

- **Histones:**

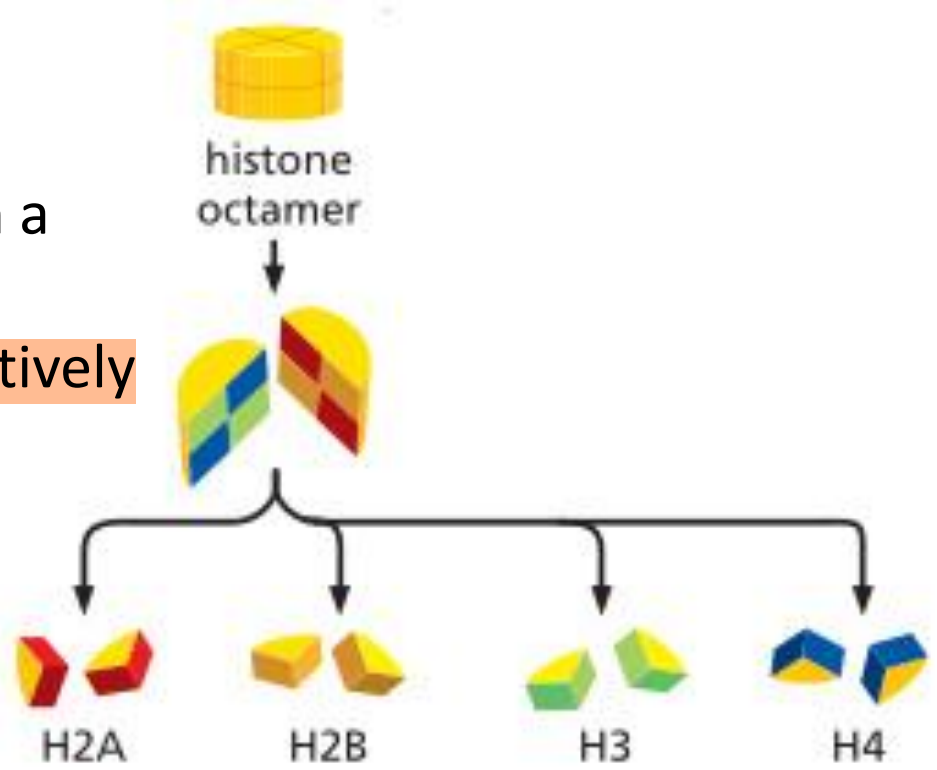
Histones are proteins that are responsible for the first level of DNA packing in chromatin.

- Histones can undergo chemical modifications that result in changes in chromatin organization. (Euchromatin (loosely packed allows transcription) vs heterochromatin (densely packed does not allow transcription))

– For example, phosphorylation of a specific amino acid on a histone tail affects chromosomal behavior during meiosis

– Histones are positively charged and hold tightly the negatively charged DNA.

– Four types of histones are most common in chromatin.

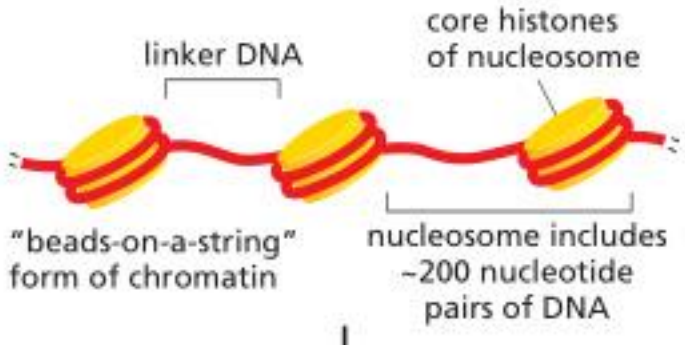


A chromosome consists of a DNA molecule packed together with proteins

• **Nucleosomes:** In electron micrographs, unfolded chromatin is roughly 10 nm in diameter (the 10-nm fiber). Such chromatin resembles beads on a string. Each “bead” is a nucleosome, the basic unit of DNA packing; the “string” between beads is called linker DNA.

A nucleosome consists of DNA wound twice around a protein core of eight histones, two each of the four main histone types. The amino end of each histone (the histone tail) extends outward from the nucleosome and is involved in regulation of gene expression.

In the cell cycle, the histones leave the DNA only briefly during DNA replication. Generally, they do the same during the process of transcription, which requires access to the DNA by transcription proteins.

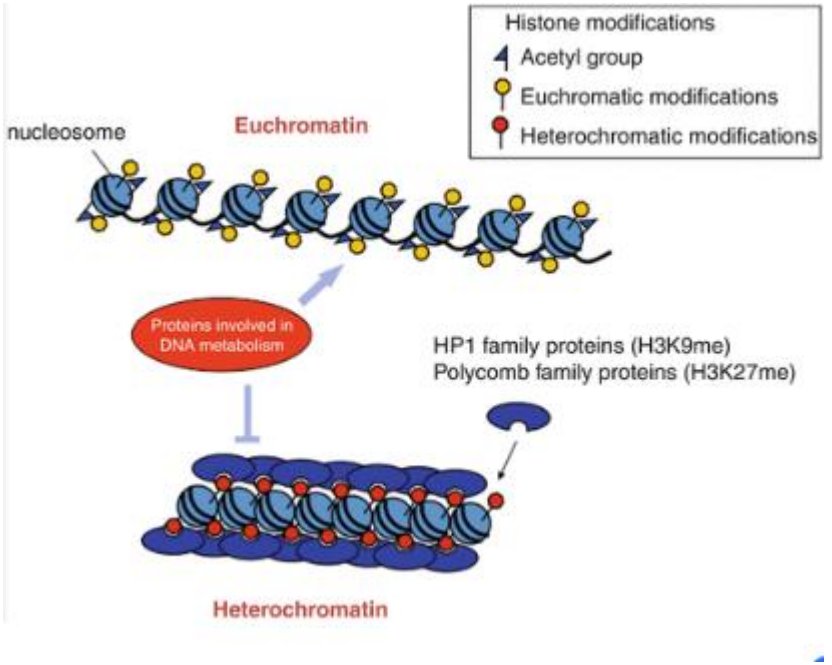


Heterochromatin Is Highly Condensed and Restricts Gene Expression

• Most chromatin is loosely packed in the nucleus during interphase and condenses prior to mitosis. Loosely packed chromatin is called **euchromatin.**

• During interphase a few regions of chromatin (centromeres and telomeres) are highly condensed into **Heterochromatin.**

• Dense packing of the heterochromatin makes it difficult for the cell to Express genetic information coded in these regions



A chromosome consists of a DNA molecule packed together with proteins

- The chromatin of each chromosome occupies a specific restricted area within the nucleus, preventing tangling of the chromatin fibers of different chromosomes.

- Even during interphase, the centromeres and telomeres of chromosomes exist in a highly condensed state similar to that seen in a metaphase chromosome.

- This type of interphase chromatin is called heterochromatin. Because of its compaction, heterochromatin DNA is largely inaccessible to the machinery in the cell responsible for expressing the genetic information coded in the DNA.

- More loosely packed euchromatin makes its DNA accessible to this machinery, so the genes present in euchromatin can be expressed.

A chromosome consists of a DNA molecule packed together with proteins

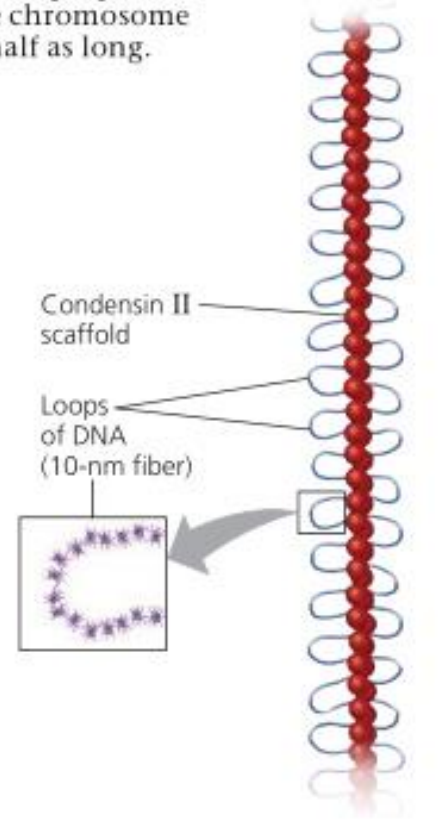
Early on, biologists assumed that interphase chromatin was a tangled mass in the nucleus but this is not the case.
Although an interphase chromosome lacks an obvious scaffold, there are proteins that further organize the 10-nm fiber into larger compartments and smaller looped domains.
Some of the looped domains appear to be attached to the nuclear lamina, on the inside of the nuclear envelope, and perhaps also to fibers of the nuclear matrix.

These attachments may help organize regions of chromatin where genes are active.
The chromatin of each chromosome occupies a specific restricted area within the interphase nucleus, and the chromatin fibers of different chromosomes do not appear to be entangled

As a cell prepares for mitosis, its chromatin becomes organized into loops and coils, eventually condensing into a characteristic number of short, thick metaphase chromosomes that are distinguishable from each other with the light microscope

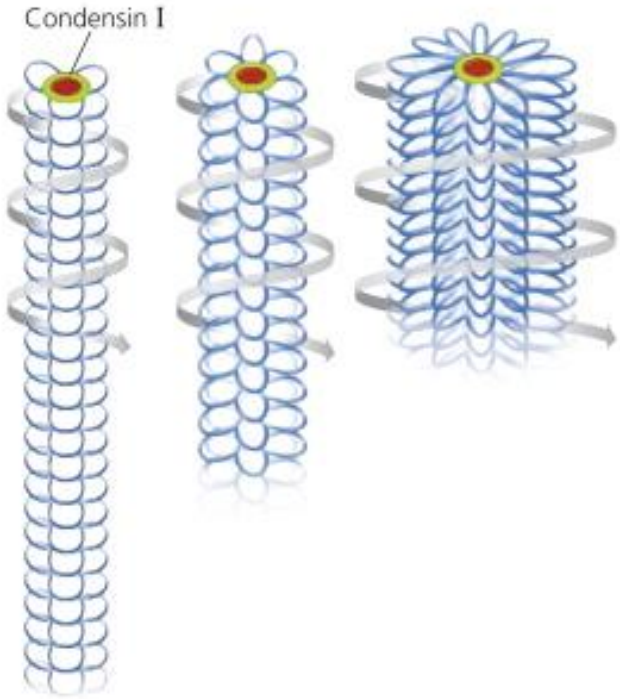
Prophase

When mitosis begins, DNA replication has already occurred, so each chromosome consists of two sister chromatids. During prophase, the chromatin of each sister chromatid begins to condense. Two related proteins called condensin II and condensin I play important roles. First, condensin II proteins (red) bind to the 10-nm fiber of DNA and form DNA loops that get larger and larger. The condensin II proteins form a central scaffold from which the loops extend. As the loops grow, the chromosome gets wider and shorter. By the end of prophase, the chromosome is half as long.



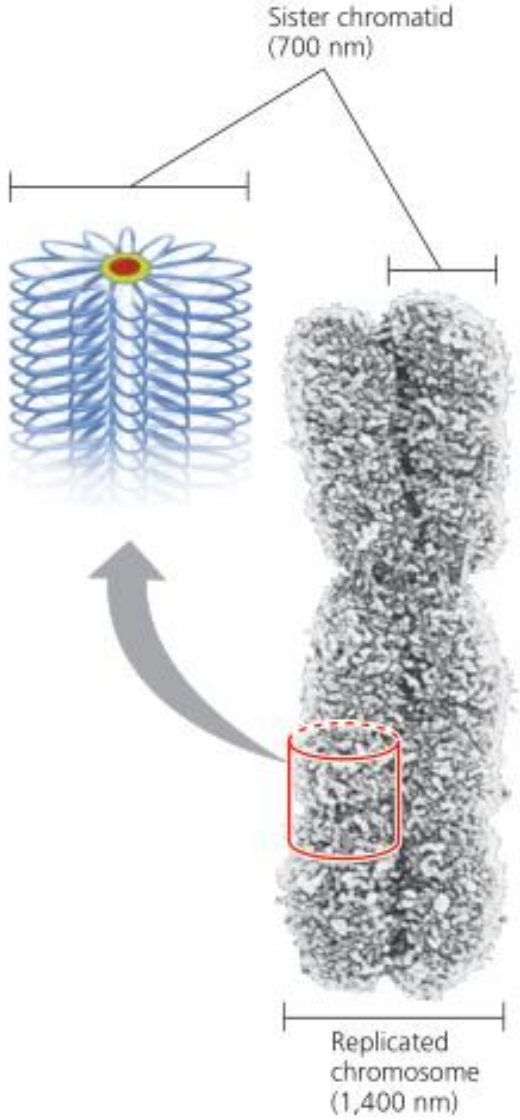
Prometaphase

When prometaphase begins, condensin I proteins (green) bind to DNA outside the central scaffold, making smaller loops (not shown) out of the larger loops generated by condensin II. The process continues, with more and more loops extending outward, and the chromosome getting denser, shorter, and wider. The scaffold itself also begins to twist into a helix (suggested by the curved gray arrows), allowing even more loops per given length of chromosome.



Metaphase

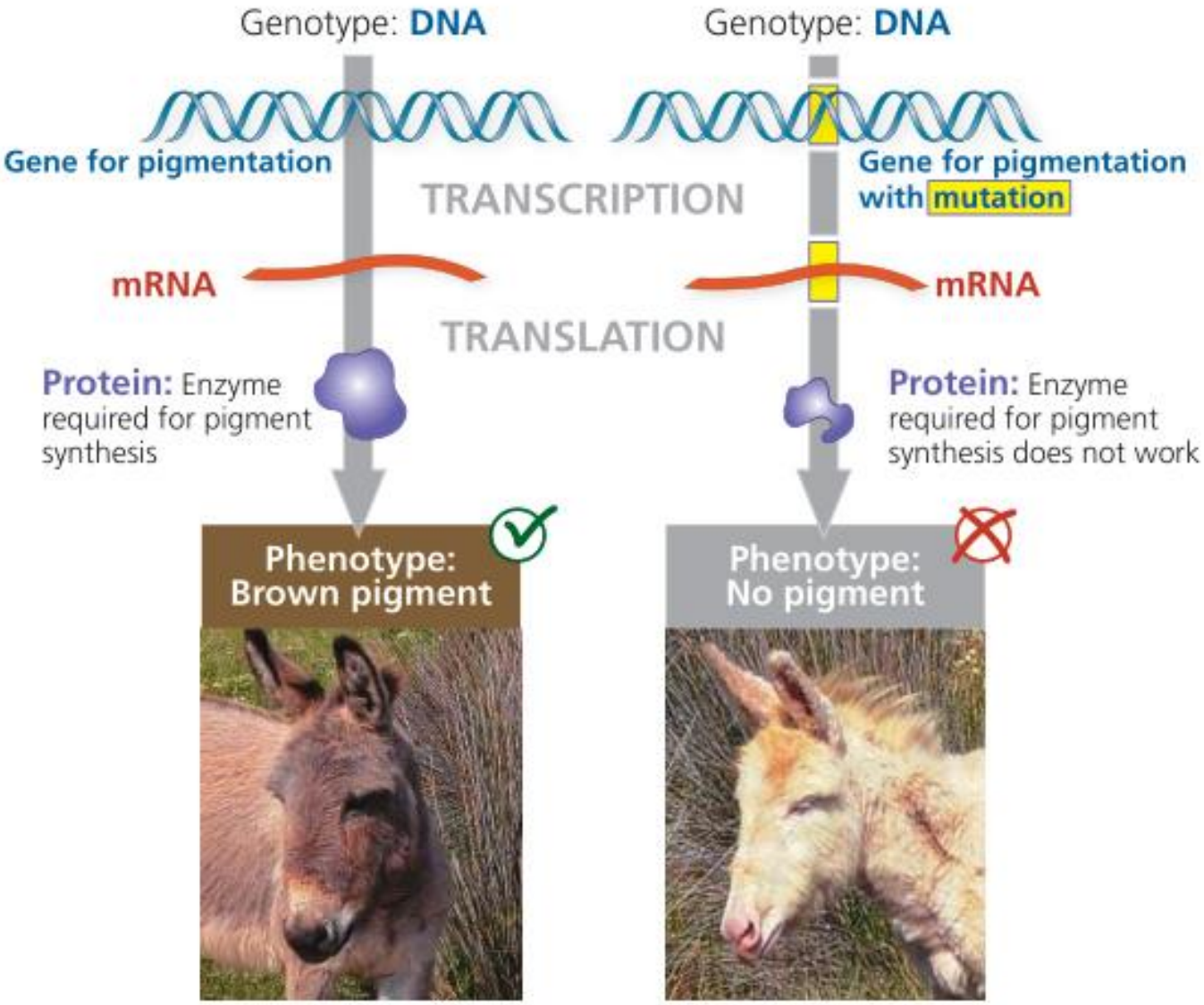
At metaphase, the chromosome is at its most dense, with the most loops per turn, and therefore is at its shortest length. The two sister chromatids are fully condensed.



Gene Expression

How can one change in DNA result in such a dramatic change in appearance?

Proteins are the link between genotype and phenotype. Gene expression is the process by which DNA directs the synthesis of proteins:



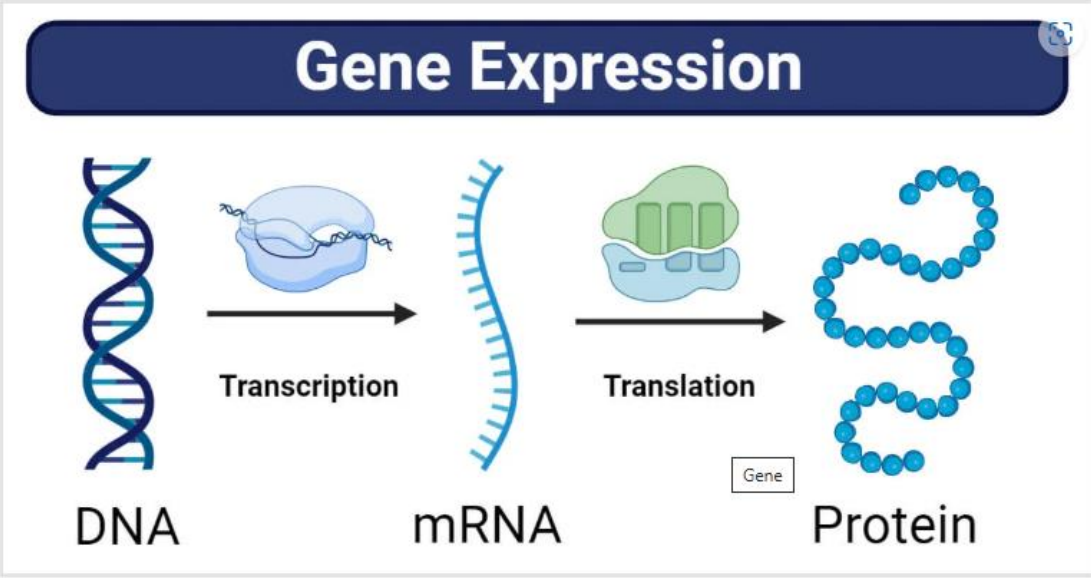
Gene Expression: From Gene To Protein

The information in DNA is in the form of specific sequences of nucleotides.

The DNA inherited by an organism leads to specific traits by dictating the synthesis of proteins.

Proteins are the links between genotype and phenotype.

Gene expression is the process by which DNA directs the synthesis of proteins and sometimes RNA's.



Gene Expression: From Gene To Protein

• In 1902, British physician Archibald Garrod was the first to suggest that genes dictate phenotypes through enzymes (proteins that catalyze specific chemical reactions in the cell).

• He thought symptoms of an inherited disease reflect an inability to synthesize a certain enzyme.
He referred to such diseases as “inborn errors of metabolism.”

• Several decades later, research supported Garrod’s hypothesis that a gene dictates the production of a specific enzyme, later named the **one gene–one enzyme** hypothesis.

• Linking genes to enzymes required understanding that cells synthesize and degrade most organic molecules via metabolic pathways, in which each chemical reaction in a sequence is catalyzed by a specific enzyme.

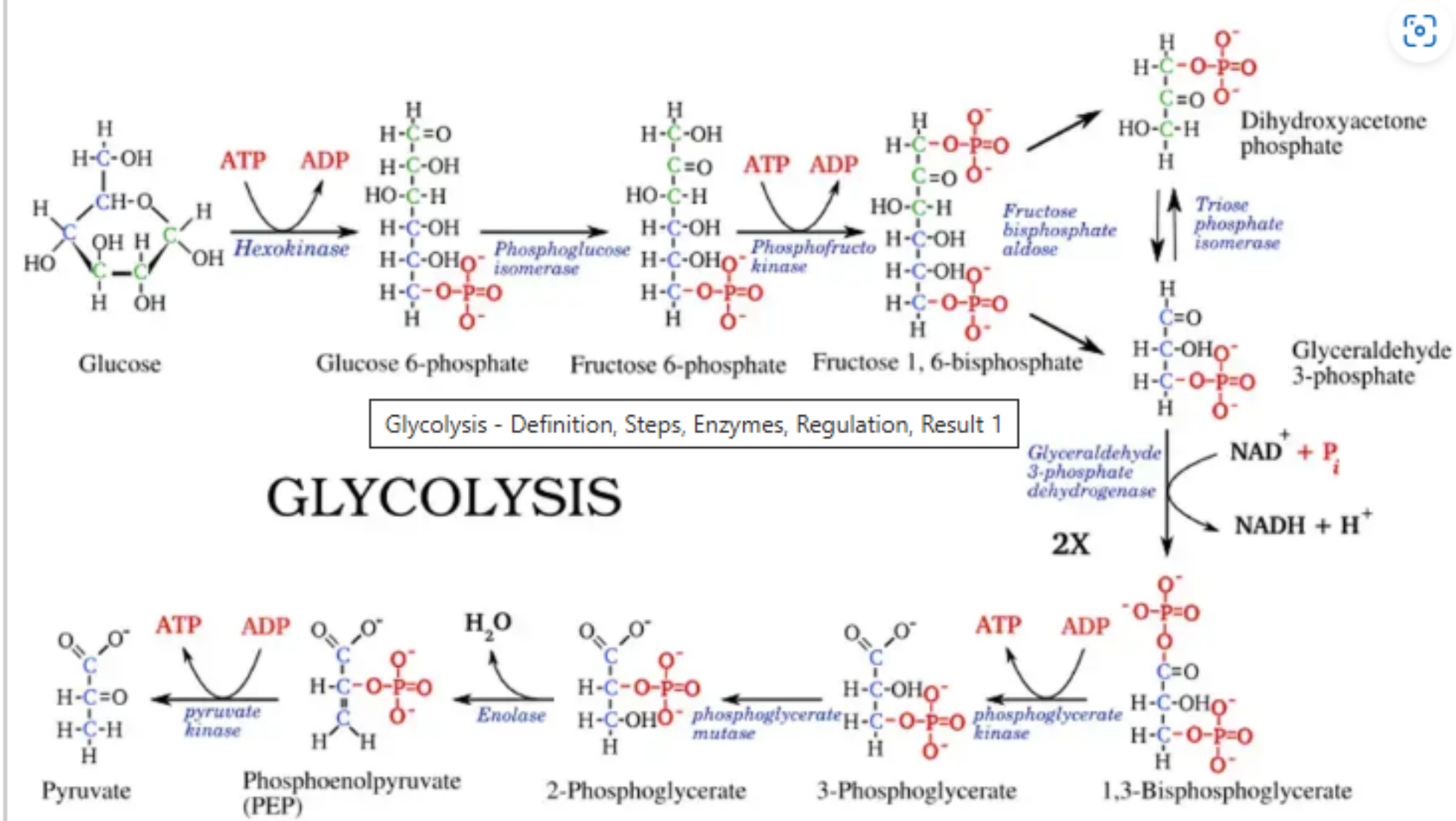


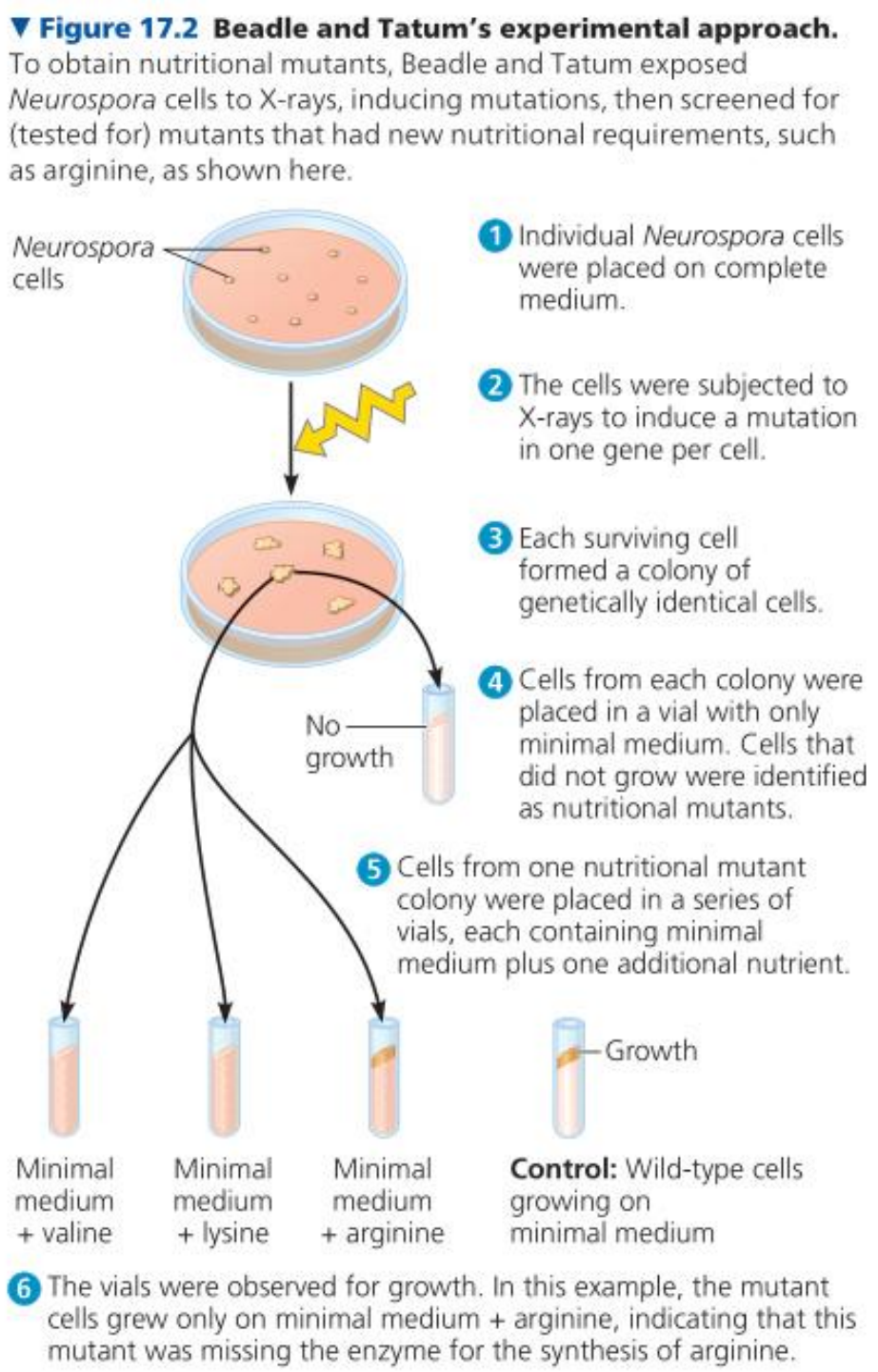
Fig: Glycolysis Pathway: Definition, Steps | *Image Source: https://miro.medium.com/max/700/0*R6_3FLZHGIAaLXJA.png*

Nutritional Mutants in Neurospora:

Evidence that demonstrates the relationship between genes and enzymes came form the study of Beadle and Tatum who experimented with bread mold, Neurospora crassa.

Neurospora is a **haploid** species unlike Mendel's peas and Morgan's flies. It only contains one copy of each chromosome in its cells. Therefore, a change in genotype is reflected in the phenotype because only one single copy of each gene determines the phenotype.

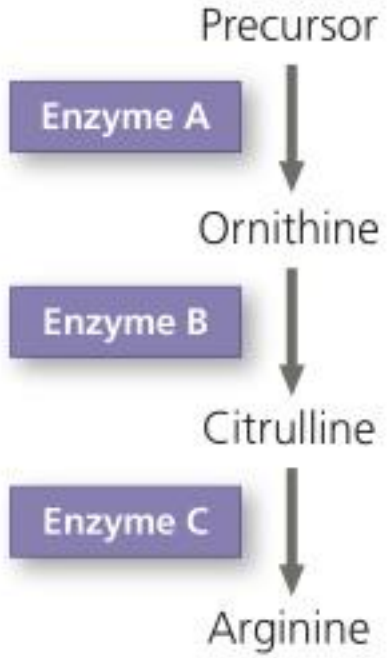
Beadle and Tatum created random mutations on the neurospora by exposing them to UV radiation and tried to disable genes that are required for a certain nutritional activity.



▼ **Figure 17.2 Beadle and Tatum's experimental approach.** To obtain nutritional mutants, Beadle and Tatum exposed *Neurospora* cells to X-rays, inducing mutations, then screened for (tested for) mutants that had new nutritional requirements, such as arginine, as shown here.

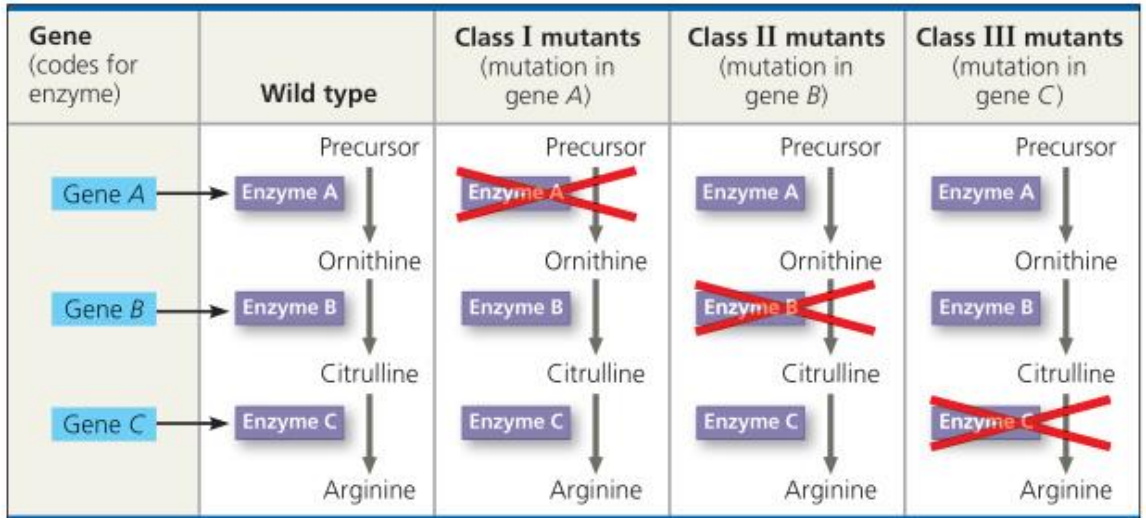
Nutritional Mutants in Neurospora:

Another research team Adrian Srb and Norman Horowitz isolated 3 different Neurospora mutants that required arginine in their growth medium. They showed that each of these three classes of mutants were defective in a different gene.
They suspected that the metabolic pathway of arginine biosynthesis involved a precursor nutrient and the intermediate molecules ornithine and citrulline.



They tested both the one gene–one enzyme hypothesis and their postulated arginine-synthesizing pathway. In this experiment, they grew their three classes of mutants under the four different conditions shown in the results table below. They included minimal Medium as a control, knowing that mutant cells could not grow on minimal medium.

Results Table	Classes of *Neurospora crassa*			
Condition	Wild type	Class I mutants	Class II mutants	Class III mutants
Minimal medium (MM) (control)				
MM + ornithine				
MM + citrulline				
MM + arginine (control)				
Summary of results	Can grow with or without any supplements	Can grow on ornithine, citrulline, or arginine	Can grow only on citrulline or arginine	Require arginine to grow



The wild-type strain was capable of growth under all experimental conditions, requiring only the minimal medium. The three classes of mutants each had a specific set of growth requirements. For example, class II mutants could not grow when ornithine alone was added but could grow when either citrulline or arginine was added.

Because each of their mutants was mutated in a single gene, they concluded that each mutated gene must normally dictate the production of one enzyme. Their results supported the one gene–one enzyme hypothesis, proposed by Beadle and Tatum, and also confirmed that the arginine pathway operates in Neurospora.

The Products of Gene Expression: A Developing Story

• Some proteins aren't enzymes (such as microtubules or insulin), so researchers later revised the hypothesis: one gene–one protein

• Many proteins are composed of several polypeptides (such as hemoglobin which consists of 2different classes of polypeptides), each of which has its own gene

• Therefore, Beadle and Tatum's hypothesis is now restated as the one gene–one polypeptide hypothesis

• Yet the name can still change because there are genes which code for RNA molecules that have important functions in cells even though they are never translated into protein.

Basic Principles of Transcription and Translation

• Genes provide the instructions for making specific proteins, but a gene does not build a protein directly.

• RNA is the intermediate between genes and the proteins for which they code

• RNA is chemically similar to DNA except that it contains ribose instead of deoxyribose and has the nitrogenous base uracil rather than thymine.

•The expression of genes that code for proteins includes two stages: transcription and translation.

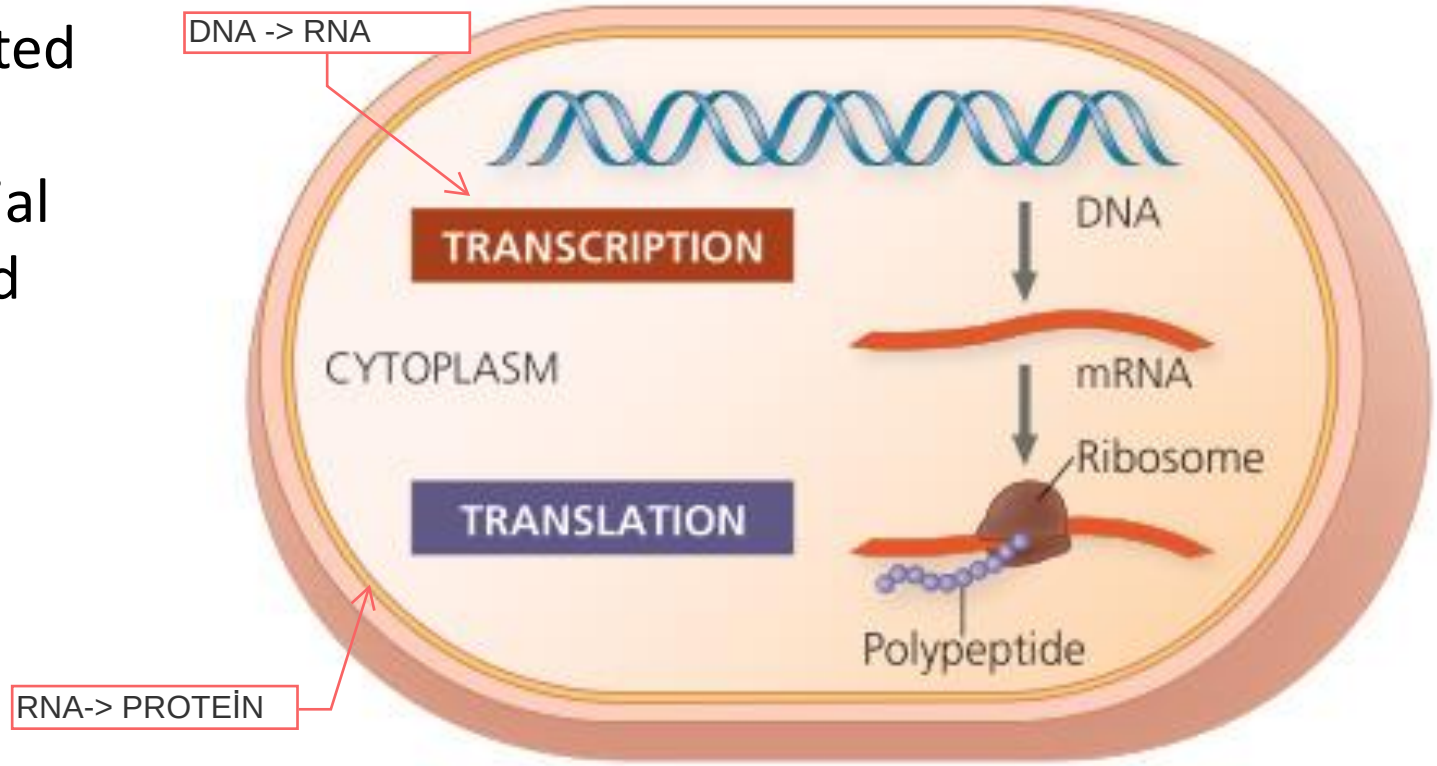
Transcription is the synthesis (production) of messenger RNA (mRNA) using the information in the DNA.

Translation is the synthesis of a protein (polypeptide) using the information in the mRNA.

Ribosomes are the sites of translation.

Basic Principles of Transcription and Translation

•In prokaryotes, mRNA produced by transcription is immediately translated without more processing. Nuclear membranes do not separate bacterial DNA and mRNA from ribosomes and the other protein-synthesizing equipment.



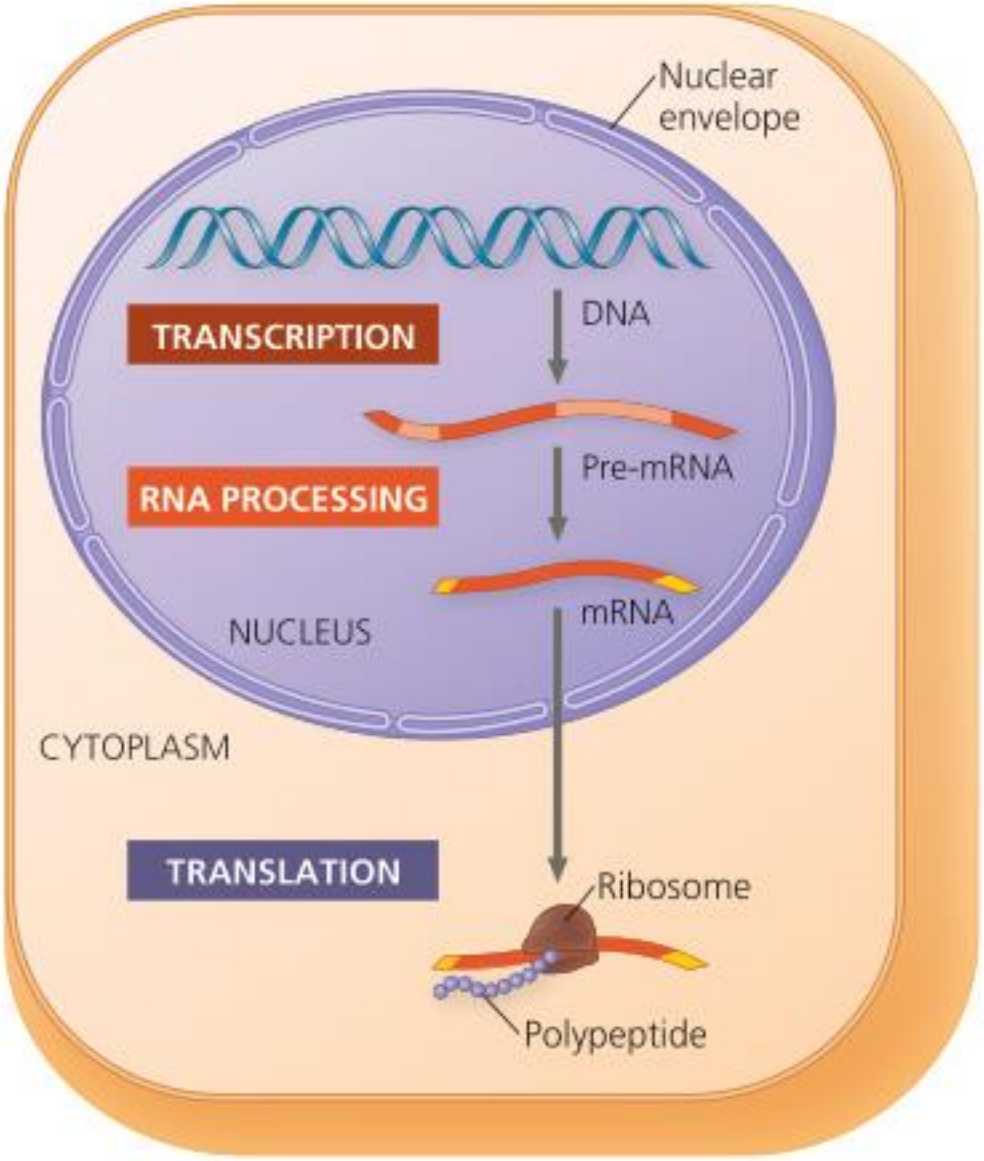
(a) Bacterial cell. In a bacterial cell, which lacks a nucleus, mRNA produced by transcription is immediately translated without additional processing.

•In a eukaryotic cell, the presence of a nuclear envelope separates transcription from translation in space and time.

•Transcription occurs in the nucleus, but the mRNA must be transported to the cytoplasm for translation.

•In eukaryotes, before RNA transcripts from protein-coding genes can leave the nucleus, they are modified in various ways to produce the final, functional mRNA.

•The transcription of a protein-coding eukaryotic gene results in pre-mRNA, and further RNA processing yields the finished mRNA.

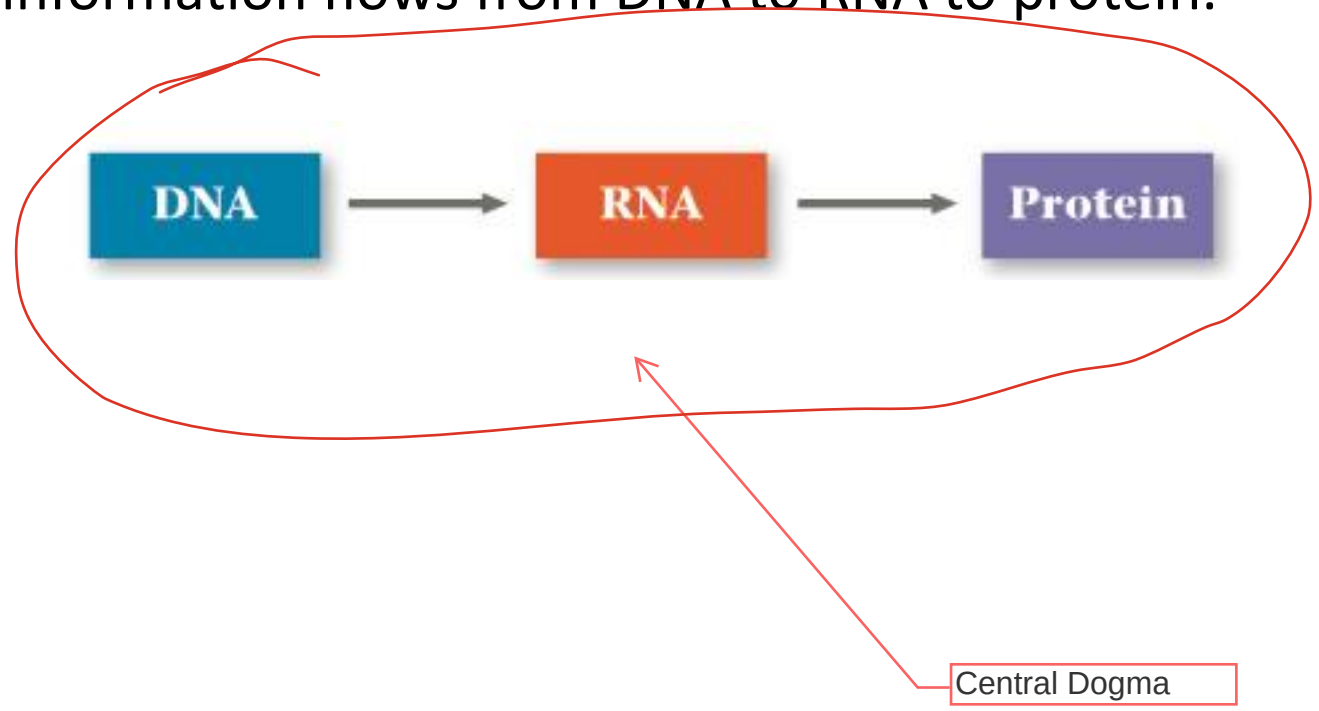


(b) Eukaryotic cell. The nucleus provides a separate compartment for transcription. The original RNA transcript, called pre-mRNA, is processed in various ways before leaving the nucleus as mRNA.

The central dogma

•Genes program protein synthesis via genetic messages in the form of messenger RNA.

•The central dogma is the concept that cells are governed by a cellular chain of command: in general, genetic information flows from DNA to RNA to protein.



The Genetic Code

- In transcription it is as if a message written out by hand is being converted, into a typewritten text. The language itself and the form of the message do not change, and the symbols used are closely related.

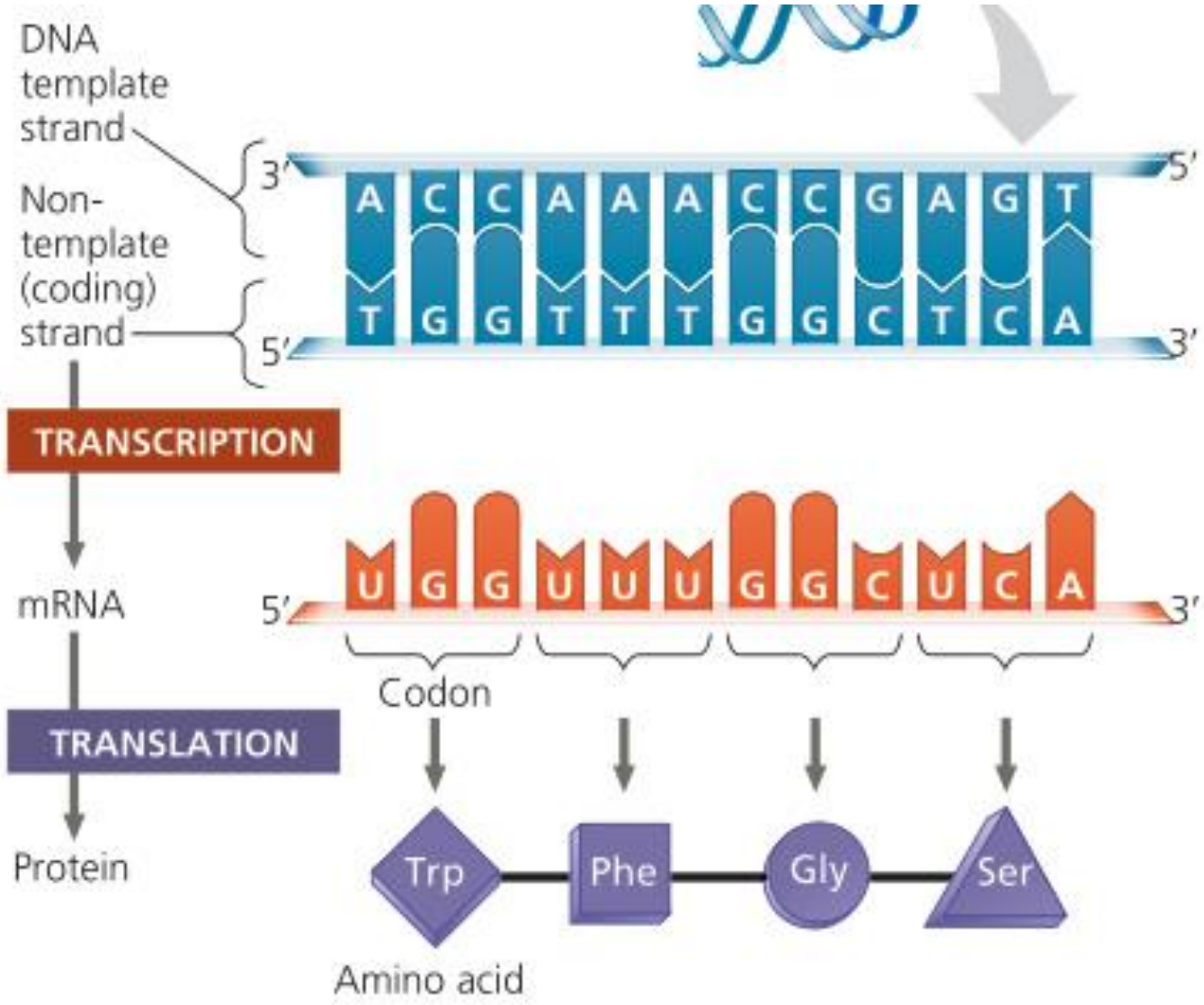
- In contrast, the conversion of the information in RNA into protein represents a translation of the information into another language that uses quite different symbols.

- The cell must translate the nucleotide sequence of mRNA molecules into amino acid sequences of proteins.

- Because there are only 4 different nucleotides in mRNA and 20 different types of amino acids in a protein, this translation cannot be accounted for by a direct one-to-one correspondence between a nucleotide in RNA and an amino acid in protein.

- How many nucleotides then should correspond to a particular amino acid.

The Genetic Code



- Experiments have verified that the flow of information from gene to protein is based on a triplet code

- Example: AGT at a particular position on a DNA strand results in the placement of the amino acid serine at the corresponding position of the polypeptide to be produced

The Genetic Code

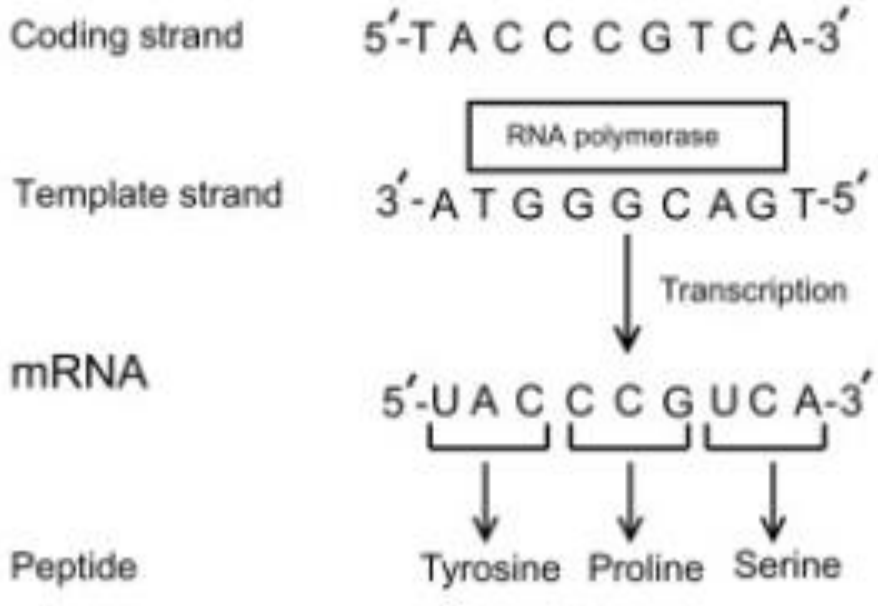
For each gene, only one of the two DNA strands is transcribed. This strand, the **template strand** provides a template for ordering the sequence of nucleotides in an RNA transcript.

Notice that an mRNA molecule is complementary rather than identical to its DNA template. It is identical to the non-template strand instead. The non-template strand is called the **coding strand**.

During translation, the mRNA base triplets, called **codons**, are read in the 5′ to 3′ direction along the mRNA.

Each codon specifies the amino acid to be placed at the corresponding position along a polypeptide.

Each codon specifies which one of the 20 amino acids will be incorporated at the corresponding position along a polypeptide.



The codon table for mRNA

All 64 codons were deciphered by the mid-1960s

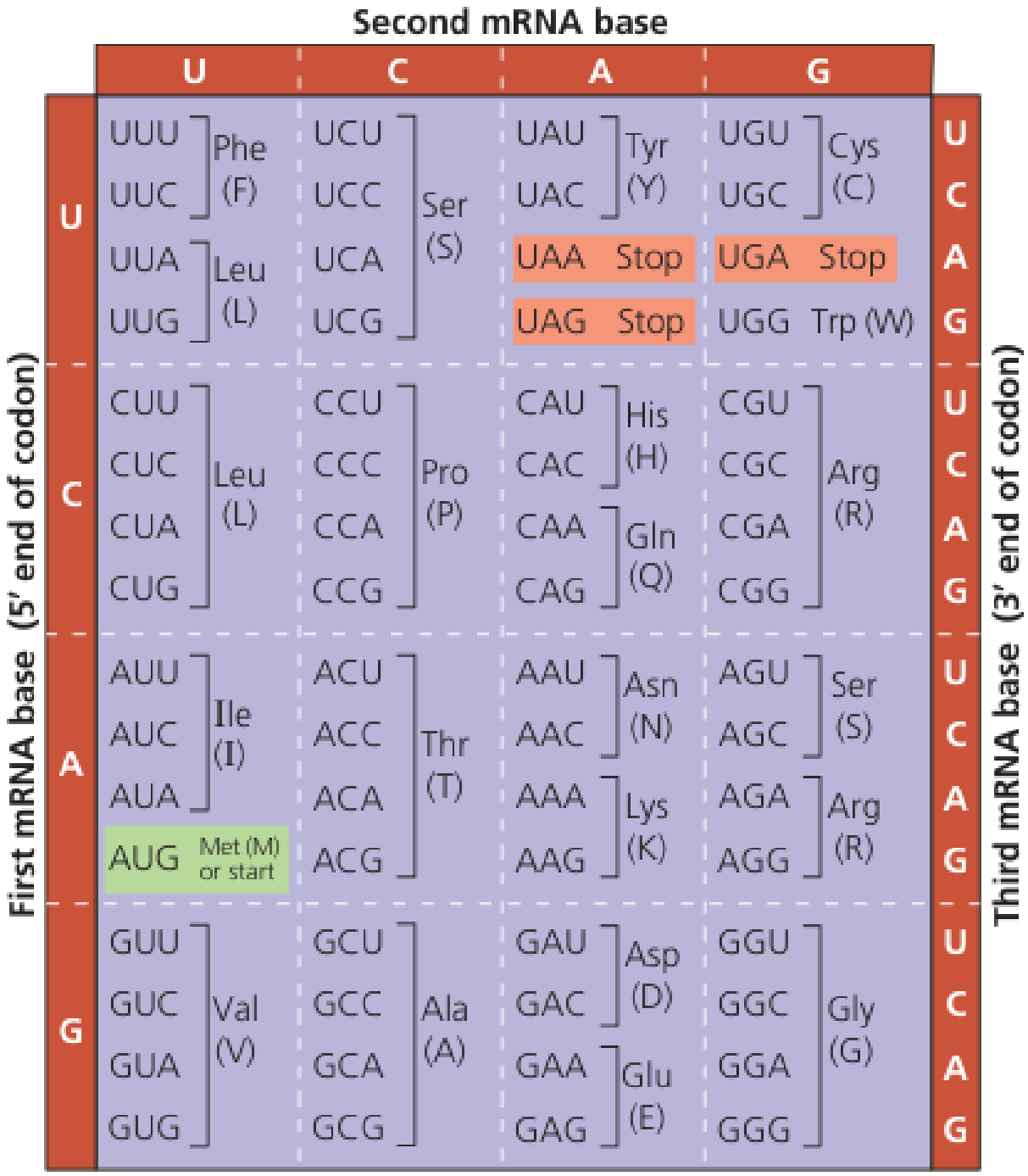
Of the 64 triplets, 61 code for amino acids; 3 triplets are “stop” signals to end translation

The genetic code is redundant but not ambiguous; no codon specifies more than one amino acid

Codons must be read in the correct reading frame (correct groupings) in order for the specified polypeptide to be produced

Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
GCA	AGA						GGA			UUA				CCA	AGC				GUA	
GCC	AGG						GGC		AUA	UUG				CCC	AGU				GUC	
GCG	CGA						GGG		AUC	CUA				CCG	UCA	ACA			GUG	UAA
GCU	CGC						GGU		AUU	CUC				CCU	UCC	ACC			GUU	UAG
	CGG	GAC	AAC	UGC	GAA	CAA		CAC		CUG	AAA		UUC		UCG	ACG		UAC		UGA
	CGU	GAU	AAU	UGU	GAG	CAG		CAU		CUU	AAG	AUG	UUU		UCU	ACU	UGG	UAU		

The codon table for mRNA



Evolution of the Genetic Code

The genetic code is nearly universal, shared by the simplest bacteria to the most complex animals

Genes can be transcribed and translated after being transplanted from one species to another

▼ Figure 17.7 Evidence for evolution: expression of genes from different species. Because diverse forms of life share a common genetic code due to their shared ancestry, one species can be programmed to produce proteins characteristic of a second species by introducing DNA from the second species into the first.



(a) Tobacco plant expressing a firefly gene. The yellow glow is produced by a chemical reaction catalyzed by the protein product of the firefly gene.



(b) Mosquito larva expressing a jellyfish gene. The gene codes for green fluorescent protein (GFP) and is inserted into organisms as a *reporter gene* so researchers can tell if a gene of interest is being expressed.